



Archaeometallurgical Characterization of Seleucid Lead Bronze Coins: Towards an Understanding of Flans Standardization

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Abstract

Five groups of copper-based coins from the Hellenistic (Seleucid) site of Tell Izṭabba (Israel), dated to the second century BCE, are the subject of this study. Various analyses were applied to study the coins' composition, microstructure, and manufacturing process and to evaluate whether a centralized workshop produced the flans regionally or mints were responsible for their production. Our results show that the coins were made of copper-tin-lead alloys except for one coin. The high lead content likely reflects economic considerations and technological requirements, suggesting that the coins were minted and struck in the same workshops. LIA data indicate that Levantine mints likely utilized similar ore sources even under Seleucid rule. Allocation to raw material sources is hampered by several factors. This study provides new insights into the technology and economy of bronze coin production in the southern Levant during the Hellenistic period, shedding light on the region's material culture.

Keywords Archaeometallurgy · Coins · Copper-tin-lead alloy · Materials characterization · Microstructure · Tell Izṭabba

Introduction

The German-Israeli Tell Izṭabba Excavation Project (GITIEP) was initiated with the purpose of investigating the short-lived Seleucid settlement of Nysa-Scythopolis/Beth She'an, located on Tell Izṭabba [1, 2]. Seleucid Nysa-Scythopolis was founded in the time of Antiochos IV in the 160 s BCE and destroyed by the Hasmonaeen John Hyrcanus I in spring/summer of 107 BCE [3]. The late Hellenistic and Roman reoccupation of the settlement took place in an adjacent area, to the south of the foot of the biblical Tell Beth She'an, evidencing that the Seleucid site had been left in

ruins and only sparsely settled in later times. A notable reoccupation of Tell Izṭabba took place during the Byzantine period, when the northern flank of the city wall crossed the entire mound from east to west and ecclesiastical buildings were erected on Tell Izṭabba's western parts.

From a numismatic perspective, Tell Izṭabba is well covered in the literature. Coins from the site's previous excavations by the Israel Antiquities Authority were recently published [4], and those from the different campaigns of the GITIEP were published as well, normally after the end of each season [5–8]; with those of the 2025 and 2026 seasons still pending publication. All these published coin finds, numbering several hundred altogether, mirror the site's settlement history.

In the lines below, we aim to address a somewhat overlooked primary research question: to what extent royal mints of the Seleucid Levant have used “standardized” copper alloy flans of identical (or similar) chemical composition in accordance with a “formal” recipe, or alternatively, whether flans were made independently by the mints while minting their coins? Obviously to answer this question in a satisfactory manner one should analyse many coins in a systematic manner, namely series of different mints and similar dates. Our below attempt is a preliminary trial to answer this research question, as we have analysed coins of five groups,

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ranging in dates in the 130 s and 120 s BCE (with the exception of Group V of the 150 s BCE), assigned to the mints of Antioch on the Orontes (the Seleucid's empire capital), as well as Aké-Ptolemaïs and Nysa-Scythopolis, and to different kings, with a focus on Alexander Zabinas (128–123 BCE) but also his predecessor Antiochos VII (138–129 BCE) and his successor Antiochos VIII (125–96 BCE). A secondary research question concerns the supply chain, including organization of metal supply to the mints and how this may have influenced the observed alloy compositions. In particular, were mints expected to procure copper, tin, and lead locally, were these metals distributed from a central authority, or were pre-alloyed Cu-Sn-Pb ingots supplied to the mints? A comprehensive answer to this question would require a larger collection of coins from multiple mints and closely dated series in order to detect patterns consistent with different supply models, which goes beyond the scope of the present study.

Metallurgical Background to Research

Metallurgical study of copper alloy coins is of great importance since it may provide information concerning the daily life, political considerations, economic crisis, as well as technological abilities of ancient civilizations [9, 10]. During the solidification stage of cast copper alloy items, the solute alloying elements and impurities tend to redistribute within the solidified alloy [11]. Therefore, cast copper items, such as coins, are frequently heterogeneous in composition and microstructure [12].

During antiquity, copper was usually alloyed with tin, zinc, and lead. Any other metallic elements detected in minute amounts were often unintended contaminants [11, 13]. Although pure copper is susceptible to work-hardening during mechanical deformation, and its hardness can increase significantly through repeated hammering [14], copper-tin and copper-tin-lead alloys were generally preferred for coin production in antiquity [12]. The presence of tin, zinc and lead as major, minor or trace elements in copper alloys could result from the composition of the raw materials or have been deliberately added to the copper to improve the properties of the item [11, 15]. Ancient coppersmiths were able to create the desired alloy (i.e. control the amount of tin) needed to produce bronze suitable for specific applications [16, 17]. By adding tin to molten copper, the melting temperature reduces from 1084 to 950 °C, resulting in a more controllable casting process. Tin also tends to reduce the amount of gas bubbles formation in the cast metal [18]. Therefore, bronzes have enhanced casting qualities compared to pure copper, with better corrosion resistance, greater hardness, as well as the ability to be further hardened by hammering [19–21].

A copper alloy with more than 2 wt.% Sn is defined as bronze in modern terminology. Although ancient bronzes could contain up to about 15 wt.% Sn, such high tin levels were uncommon. The presence of tin in bronzes greatly improves the hardness and resistance to corrosion [22]. While increased tin content improved the alloy's castability, it also significantly reduced the ductility of the final object, which often made it unsuitable for functional use [23, 24]. Hence, bronzes with more than 10 wt.% Sn usually remained in their as-cast condition, with no additional hammering and annealing operations [23, 25]. Low-tin bronze coins, containing less than 2.0 wt.% Sn, are typically attributed to extensive use of scrap metal, since repeated melting promotes tin oxidation and consequently lowers its concentration in the alloy [26]. Low-tin bronzes, attained from recycled objects, were used during antiquity since they were less expensive than rich-tin bronzes [23, 25].

When the tin content exceeded 6 wt.%, the alloy microstructure consisted of an α solid solution and an $(\alpha + \delta)$ eutectoid. During casting, molten metal may migrate to the surface through interdendritic channels, forming a layer that develops the eutectoid microstructure upon cooling [27]. In Cu-Sn-Pb ternary alloys, similar lamellar structures can form, with the added complexity of lead distribution. This eutectoid microstructure can sometimes be observed directly in backscattered electron (BSE) images without chemical etching, depending on the composition and contrast between the phases. Heat treatment, corrosion, or surface reactions can further modify the layers, complicating interpretation of the mechanism by which the high-tin surface was formed [28, 29].

The addition of lead to copper alloys has both metallurgical and economic implications. Bronzes with more than 1 wt.% Pb are termed “leaded bronzes” in modern terminology [29]. Lead improves the casting process by lowering the liquidus temperature and expanding the freezing range, which facilitates handling and shaping of the alloy. During antiquity, lead was often intentionally added to copper alloys to raise the fluidity (where only 2 wt.% Pb is needed to achieve optimal fluidity), enhance castability, and to attain better surface finish of the cast alloy [10, 22, 30]. High lead content was perhaps also used to improve the striking ability during the stamping of the coins [31]. In addition, lead was a cheap metal that was occasionally added to copper alloys for economic reasons [23, 32]. Yet, the economic impact of adding lead to ancient copper-based coins may have been probably less related to the cost of the metal itself and more associated with reducing labour requirements during the melting of the alloy. Since the solubility of lead in copper is extremely low (0.007 wt.% Pb), lead-rich bronzes are characterized by heterogeneous microstructure. During solidification, copper containing dissolved tin solidifies first, forming α -Cu(Sn) dendrites. As solidification proceeds, tin enriches

the remaining liquid and may precipitate along grain boundaries once its solubility limit in copper is exceeded. Lead, being practically insoluble in copper, solidifies last and segregates as globular inclusions within the interdendritic regions [12, 33]. In Cu-Sn-Pb coin alloys, lead-rich regions can be observed as bright phases in BSE imaging without chemical etching, due to the strong atomic number contrast between lead and the surrounding copper-tin matrix [23].

During the Roman (and Byzantine) period, coins made of bronzes with high lead content were more frequently used in the eastern provinces [34]. Archaeometallurgical analysis of two copper alloy coins, dated between 450 and 550 AD, retrieved from the Byzantine Dor 2006 shipwreck (Israel), revealed that the items were made of copper-lead alloy, with a heterogeneous microstructure according to scanning electron microscopy (SEM) observation and a high concentration of lead, between 32.6 and 43.8 wt.% Pb and only up to 0.7 wt.% Sn according to X-ray fluorescence (XRF) analysis. The Vickers (HV) microhardness (25 gf load) results of the two coins retrieved from Dor 2006 shipwreck revealed an average value of 97 ± 8 HV. The flans of these two coins were produced by casting, and were later struck with coin dies [23]. Archaeometallurgical analysis of various copper-based coins retrieved from the Byzantine-period Ma'agan Mikhael B shipwreck, dated to the fourth century AD based on the portrait of the House of Constantine I, revealed that the coins were made of ternary Cu-Sn-Pb alloy with a high concentration of lead (between 0.3–65.7 wt.% Pb) and up to 11.5 wt.% Sn. The average composition of the coins was 83.0 ± 7.7 wt.% Cu, 2.6 ± 1.1 wt.% Sn, and 9.4 ± 6.6 wt.% Pb, as well as minor amount of other element (based on 36 measurements obtained by XRF analysis) [35]. Metallographic observations of the coins from Ma'agan Mikhael B by SEM revealed a heterogeneous microstructure with equiaxed α -copper grains and the presence of some annealing twins, as well as preferred oriented lead participations, appearing as bright areas in BSE mode. Based on visual examination and SEM observation, the coins from the Ma'agan Mikhael B shipwreck were made of cast flans that were heated before being struck [35, 36].

Lead segregation during solidification of copper alloys produces an inherently heterogeneous distribution within the alloy. In burial environments, lead is particularly susceptible to corrosion, and the preferential alteration of lead-rich regions may locally affect the measured composition of the alloy, especially in surface-sensitive analytical techniques [23, 35, 36]. Examining the internal distribution and preservation state of lead islands can therefore be important for assessing bulk composition of ancient coins made of Cu-Sn-Pb alloys. Non-destructive X-ray microscopy, such as micro-CT analysis, allows three-dimensional visualization of dense lead inclusions within the alloy and can reveal their spatial distribution throughout the coin

without sampling damage [9]. However, its three-dimensional resolution may be limited compared to direct microstructural examination. Alternatively, destructive metallographic analysis, combined with microscopic analysis, allows detailed observation of the alloy's microstructure, including phase distribution, grain characteristics, lead islands distribution, defects, plastic deformation, and alloy heterogeneity, as well as production-related features such as annealing or surface finishing. These observations provide valuable insight into traditional manufacturing techniques and their effects on the alloy's properties, although the method is inherently destructive and requires sectioning the object [9, 35, 36]. Both methods therefore offer complementary insights into the internal heterogeneity of archaeological Cu-Sn-Pb alloys.

In the ancient Mediterranean, the relative value of copper, tin, and lead reflected both availability and the logistics of long-distance trade. Copper was more readily produced locally, yet smelting and refining remained labour-intensive and costly [37–39]. Tin, uncommon in the region and reliant on imports from distant sources, was particularly expensive, strongly constraining the production of tin-rich bronzes [40, 41], whereas lead, often a by-product of silver extraction, was cheaper [42–44]. While these economic factors likely influenced alloy compositions, the precise relationship between metal cost, availability, and metallurgical choice remains poorly documented, highlighting the need for caution when interpreting material selection in Hellenistic bronzes.

Although copper and its alloys have relatively good resistance to corrosion, in some environmental conditions, and especially after a long period of time, more destructive forms of corrosion can occur specific to copper-based alloys. This is known as “bronze disease” and is caused by chloride ions from the environment, such as soil or seawater [21, 45, 46]. The primary reactions between the copper alloy surface and the surrounding include the formation of cuprite (Cu_2O), tenorite (CuO), cassiterite (SnO_2), and malachite [$\text{Cu}_2(\text{CO}_3)(\text{OH})_2$], where cuprite oxide acts as a protective film preventing further corrosion; however, cassiterite acts as a brittle protective layer [20]. A variety of environmental factors (such as pH, oxygen content, humidity, and soluble salts content) affect the corrosion process of bronze alloys, leading to the formation of a surface patina, primarily composed of copper compounds. Yet, the presence of calcite (CaCO_3) is also common mineral in the patina layer [9]. In some cases, the patina contains notable concentrations of tin, lead, and their compounds [36, 47]. For example, in the case of leaded bronze alloys, notable concentrations of lead oxides, such as litharge (α - PbO), and lead chlorides, such as laurionite [$\text{PbCl}(\text{OH})$], are present in the patina cover [9]. Archaeological Cu-Sn-Pb alloys, such as those used in ancient coins, commonly exhibit compositional

heterogeneity due to both metallurgical processes and post-depositional corrosion.

The aims of our research are to identify the composition and manufacturing processes of the coins, to reveal, if possible, the origin of their raw material ores, and to compare the different groups of coins. Such an analysis will enable to understand whether different mints in the Levant have used flans of identical, similar or different metal composition while producing their coins.

Experimental Methods

The metallurgical characterization of the 22 bronze coins (Fig. 1, Table 1; Group I: coin nos. 1.1, 1.2, 1.3, 1.4; Group II: coin nos. 2.1, 2.2, 2.3, 2.4, 2.5, 2.6; Group III: coin nos. 3.1, 3.2, 3.3, 3.4; Group IV: coin nos. 4.1, 4.2; Group V: coin nos. 5.1, 5.2, 5.3; and test coin nos. T1, T2, T3 included the following test methods:

(a) Weight and dimensions measurements: The weight of the coins was measured using an analytical weighing device.

(b) Visual testing (VT) inspection was used to examine the state of preservation of the coins and to detect macroscopic details that could provide clues concerning their manufacturing processes. To expose the metal, the patina cover of the coins from all five groups (I–V) was cleaned by sandblasting. Then each of the coins was locally ground

with 240–1000 grit silicon carbide papers in a few areas to remove the thick patina cover and the thin brown film beneath and to expose the base alloy. In addition, three other coins (T1–T3) of unknown group were also locally ground with 240–1000 silicon carbide grit papers.

(c) Portable energy dispersive X-ray fluorescence (pED-XRF, hereafter referred to simply as XRF) chemical analysis of the coins was done with an OXFORD X-MET 7500 handheld tool, using a silicon drift detector (SDD) and low-energy (LE) operation mode, equipped with a 45 kV Rh target X-ray tube. The surface of each examined coin was cleaned by sandblasting and next washed with ethanol and dried prior to XRF analysis. Three measurements were done to each coin, where each measurement was performed in a bench-top stand for 30 s, with a spot size of 5 mm diameter. Such portable XRF analysis offers a rapid, non-invasive method for examining the composition of ancient bronze coins, while preserving the integrity of these valuable artifacts. Yet, light elements (such as oxygen and carbon) could not be spotted owing to instrumental limitations [35].

(d) SEM with energy-dispersive spectroscopy (EDS) analysis was performed on the coins' patina cover and on the coins' exposed (ground) metal by using an environmental SEM (ESEM) FEI Quanta 200FEG instrument in high vacuum mode and an Everhart–Thornley secondary electron (SE) detector, calibrated with standard samples from the manufacturer. Both SE and backscattered electron (BSE)



Fig. 1 The five groups of Hellenistic copper alloy coins and the additional three coins (T1–T3) of unknown group from Tell Izṭabba analysed in this paper

Table 1 Description and dimensions of the coins from Tell Iztabba

Sample No.	Group/Mint (with linked ref.)	Weight (g)	Diameter (mm × mm)	Obverse description	Reverse description	Date
Coin no. 1.1 (No. 7066)	I/Nysa-Scythopolis/Nysa-Scythopolis (cf. Farhi 2022)	4.90	17 × 12	Head of Dionysus to r.; serrated edge	$[BA\Sigma/\Lambda E Q \Sigma]$ on r., $[A\Lambda E \Xi A/N/\Delta P O Y]$ on l. Winged Tyche standing l., alathus on head, holding ship's tiller and cornucopiae	128–122 BCE
Coin no. 1.2 (No. 7526)		6.49	17 × 12			
Coin no. 1.3 (No. 7530)		5.39	17 × 12			
Coin no. 1.4 (No. 7532)		5.12	17 × 12			
Coin no. 2.1 (No. 7020)	II/Antioch on the Orontes	3.31	14 × 16	Winged bust of Eros to r.	$BA\Sigma/\Lambda E Q \Sigma$ $[A/\Lambda E \Xi A/N/\Delta P O Y]$ on l. Anchor, flukes upwards	129/128 BCE
Coin no. 2.2 (No. 7080)	II/Antioch on the Orontes	5.64	19 × 12	Diademed, radiate head of Alexander II Zabinas to r.; bevelled edge	$[BA/\Sigma/\Lambda E/\Omega \Sigma]$ on r., $[A\Lambda E \Xi A/N/\Delta P O Y]$ on l. Double cornucopia, in field control mark $\Sigma-A$	125–122 BCE
Coin no. 2.3 (No. 7368)	II/Antioch on the Orontes	4.05	19 × 12	Diademed, radiate head of Alexander II Zabinas to r.; bevelled edge	$[BA\Sigma/\Lambda E Q \Sigma]$ on r., $[A\Lambda E \Xi A/N/\Delta P O Y]$ on l. Double cornucopia	125–122 BCE
Coin no. 2.4 (No. 7375)	II/Antioch on the Orontes	4.93	20 × 12	Diademed head of Alexander II Zabinas to r.	$BA\Sigma/\Lambda E Q \Sigma$ on r., $[A\Lambda E \Xi A/N/\Delta P O Y]$ on l. Dionysus standing l., holding cantharus and thyrsus	129/128 BCE
Coin no. 2.5 (No. 7378)	II/Antioch on the Orontes	5.50	18 × 12	Diademed, radiate head of Alexander II Zabinas to r.; bevelled edge	$BA/\Sigma/\Lambda E Q \Sigma]$ on r., $[A\Lambda E \Xi A/N/\Delta P O Y]$ on l. Double cornucopia, in field control mark $[\Sigma]-A$	125–122 BCE
Coin no. 2.6 (No. 7017)	II/Antioch on the Orontes	6.58	18.5 × 12	Diademed, radiate head of Alexander II Zabinas to r.; bevelled edge	$BA\Sigma/\Lambda E Q \Sigma$ on r., $[A\Lambda E \Xi A/N/\Delta P O Y]$ on l. Double cornucopia, in field control mark $\Sigma-A$	125–122 BCE
Coin no. 3.1 (No. 7074)	III/Antioch on the Orontes	2.80	14 × 12	Head of lion to r.	$[BA\Sigma/\Lambda E Q \Sigma ANTIOX O Y]$ on r., $EYEP I E T O Y$ on l., Club	139/138 BCE
Coin no. 3.2 (No. 7078)	III/Antioch on the Orontes	5.83	18 × 12	Winged bust of Eros to r.; bevelled edge	$BA\Sigma/\Lambda E Q \Sigma ANTIOX O Y$ on r., $EYEP I E T O Y$ on l., below control mark, star, date COP (= SE 176), Isis headrest resting on two ears of grain with stems crossed above inverted crescent	137/136 BCE
Coin no. 3.3 (No. 7367)	III/Antioch on the Orontes	5.24	19 × 12	Winged bust of Eros to r.; bevelled edge	$BA\Sigma/\Lambda E Q \Sigma ANTIOX O Y$ on r., $EYEP I E T O Y$ on l., Below control mark, star, date COP (= SE 176), Isis headrest resting on two ears of grain with stems crossed above inverted crescent	137/136 BCE
Coin no. 3.4 (No. 7524)	III/Antioch on the Orontes	3.55	16 × 12	Winged bust of Eros to r.; bevelled edge	$[BA\Sigma/\Lambda E Q \Sigma ANTIOX O Y]$ on r., $[EYEP I E T O Y]$ on l., Below control mark, star, date COP (= SE 176), Isis headrest resting on two ears of grain with stems crossed above inverted crescent	139/138 BCE

Table 1 (continued)

Sample No.	Group/Mint (with linked ref.)	Weight (g)	Diameter (mm × mm)	Obverse description	Reverse description	Date
Coin no. 4.1 (No. 7379)	IV/Antioch on the Orontes	6.11	18 × 12	Radiate diademed head of Antiochus VIII to r.; bevelled edge	[BAΣ/ΛEQΣ]/ANTIOX[OY] on r., EΠIΦA/NOYΣ/ on l. Eagle standing l., sceptre under far wing	121–120 BCE
Coin no. 4.2 (No. 7528)	IV/Antioch on the Orontes	4.33	17 × 12	Diademed head of Antiochus VIII to r.; bevelled edge	BAΣ/ΛEQΣ ANTIOXOY EΠIΦANOYΣ Eagle standing l., sceptre under far wing, ΔA in l. field	113/112 BCE
Coin no. 5.1 (No. 7030)	V/Aké-Ptolemaï	5.83	18 × 12	Bust of Athena in crested Corinthian helmet to r.; bevelled edge	BAΣ/ΛEQΣ ANTIOXOY on r., EYEPΓETOY on l., in l. field ΠA monogram, date HOP (= SE 178), owl standing three quarters r., head facing	135/134 BCE
Coin no. 5.2 (No. 7075)	V/Tyre	7.04	19 × 12	Diademed head of Demetrius I to r.	BAΣ/ΛE/ΩΣ/ / ΔHMHTΠIOY in two lines above galley, below TYP/ΩN/ [-]Δ in Phoenician, date ΛONP (= SE 159) above stern	154/153 BCE
Coin no. 5.3 (No. 7377)	V/Tyre	2.97	15 × 12	Diademed head of Demetrius I to r.	[BAΣ/ΛE/ΩΣ/ ΔHMHTΠIOY] Palm tree, date NP (= SE 154) in r. field	159/158 BCE

modes were applied. The composition was detected using EDS with a silicon-lithium [Si(Li)] liquid-cooled Oxford X-ray detector. SEM-EDS analysis of bronze coins gives information about the examined surface microstructure and surface chemical composition, linking between the microstructure and composition (including X-ray elemental maps of elements distribution), as well as information related to corrosion forms [48].

(e) X-ray diffraction (XRD) analysis was performed to identify the crystalline phases of the thick patina and mud cover, as well as the crystalline phases of the thin brown film after the removal of the green patina and hard mud cover. The XRD pattern was achieved by using a Bruker's D8 Discover Diffractometer. A theta-theta (Θ - Θ) geometry setup with a Göbel Mirror was employed to produce a parallel beam, coupled with longitudinal Söller slits, for analysis. The source was a sealed X-ray tube with copper anode, with 40 kV tube voltage working condition and 40 mA tube current working condition. The detector was a LYNXEYE XE linear detector. The XRD patterns were collected between 20 and 70 (2Θ) range, with a step size of 0.04° (2Θ) for 1 second per step.

(f) Inductively coupled plasma mass spectrometry (ICP-MS) analysis from solution was performed with Element XR high-resolution instrument. The ICP-MS method allows to determine the accurate elemental composition of the coins' alloy, including major alloying elements as well as minor and trace elements. The technique, with high sensitivity and precision, was employed on solution of metallic powder drilled from several bulk areas of coin nos. 1.2, 2.6, 3.2, 4.1, 5.1, and 5.2. To minimise the problem of silver precipitation, we tested three digestion series using: (1) diluted HCl, (2) diluted HNO₃, and (3) aqua regia. In neither case were any precipitates observed. The different digestion series resulted in measured sums that had not differed from each other. Even assuming that invisible amounts of silver were lost during the dissolution process, the measured totals suggest that the maximum loss of silver would not exceed 0.002 wt.% Ag (given the 92% sum example and a resulting assumed maximum loss of 8 % AgCl), which would not significantly change the range of silver content in the samples.

In the ICP-MS method, each sample solution is introduced into an argon plasma, where the atoms are ionized at temperatures between 6,000 and 10,000 K. The resulting ions are then separated in the mass spectrometer according to their mass-to-charge ratios (m/z), where m is the ion's mass and z is its charge [49]. Internal standards are used to correct for instrumental drift and matrix effects, while calibration is performed using multi-element standard solutions of known concentrations [50]. The elemental data obtained provide insights into the metallurgical practices, technological skills, and possible recycling processes involved in the

production of the coins, and possible raw material sources [51].

(g) Lead isotope analysis (LIA) of coin nos. 1.2, 2.6, 3.2, 4.1, 5.1, and 5.2 was performed to identify possible ore sources of the lead raw metal. The isotopic data were acquired using an ICP mass spectrometer. The isotopic ratios of lead: $^{206}\text{Pb}/^{204}\text{Pb}$, $^{207}\text{Pb}/^{204}\text{Pb}$, and $^{208}\text{Pb}/^{204}\text{Pb}$ were measured and compared with reference data from known ore deposits to infer the provenance of the metal [35, 52, 53]. Due to the archaeological context, focus was laid on a comparison with ores of Fenan and Timna, Cyprus, and Anatolia.

Results

Coin Nos. 1.1, 1.2, 1.3, 1.4 (Group I = SC 2242, Alexander Zabinas, mint of Nysa-Scythopolis; cf. Farhi, 2022 [54])

Group I of the assumed mint of Nysa-Scythopolis includes four copper alloy coins, where the circumference of the coins is serrated (Fig. 1). The coins' weight ranges between 4.90 g and 6.49 g. The description of the images on the obverse and reverse of the coins is given in Table 1.

The XRF results after sandblasting cleaning indicate the compositional heterogeneity of Group I's coins (Table 2). The average composition of Group I's coins according to XRF analysis is 71.9 ± 9.1 wt.% Cu, 9.5 ± 1.9 wt.% Sn, and 18.6 ± 7.5 wt.% Pb (12 measurements of the four coins). The compositional heterogeneity of Group I's coins is illustrated in the ternary diagram of the alloy composition (Fig. 2a). The spread of data points reflects chemical differences among the coins, highlighting the non-uniform distribution of alloying elements within this group.

SEM observation of the coins exposed metal after locally grinding its surface revealed dark areas rich in copper that contain also tin and bright lead-rich areas according to BSE mode (Figs. 3a–f, 4a), as typical to Cu-Sn-Pb alloys [55]. Areas 1 and 2 of coin no. 1.2 (Fig. 3c and Fig. 3d–e, respectively) were examined by SEM-EDS analysis.

SEM-EDS elemental mapping of coin no. 1.2's obverse (area 1, Fig. 4a) revealed the presence of the elements: Cu, Pb, Sn, O, and Cl (Fig. 4b–f, bright dots). According to the elemental mapping results, areas rich in copper (Fig. 4b) also contain a higher amount of tin (Fig. 4d), whereas areas rich in lead (Fig. 4c) also contain a higher amount of chlorine (Fig. 4f).

EDS analysis results of Group I's coins alloy composition after sandblasting followed by locally grinding the surface with silicon carbide papers revealed the compositional heterogeneity of Group I's coins (Table 3). The average composition of Group I's coins according to EDS analysis

is 64.0 ± 8.4 wt.% Cu, 10.5 ± 3.5 wt.% Sn, and 25.5 ± 10.3 wt.% Pb (12 measurements of ground areas of four coins).

ICP-MS analysis of Coin no. 1.2 revealed that its bulk composition corresponds to a Cu-Sn-Pb alloy, consisting of 72.2 wt.% Cu, 5.9 wt.% Sn, and 21.2 wt.% Pb as the major alloying elements. Minor and trace elements were also detected in this coin, including minor amounts of nickel (0.13 wt.% Ni) and arsenic (0.25 wt.% As), and traces (less than 0.1 wt.%) of the elements Ag, Sb, Te, Hg, Bi, Au, S, Co, Zn, and Se (Table 4).

Coin Nos. 2.1, 2.2, 2.3, 2.4, 2.5, 2.6 (Group II = SC 2229, 2230, 2235, Alexander Zabinas, mint of Antioch)

Group II of the mint of Antioch on the Orontes includes six copper alloy coins (Fig. 1). The coins' weight ranges between 3.31 and 6.58 g. The description of the images on the obverse and reverse of the coins is given in Table 1.

The XRF results after sandblasting cleaning indicate the compositional heterogeneity of Group II's coins (Table 2). The average composition of Group II's coins according to XRF analysis is 69.9 ± 7.1 wt.% Cu, 9.7 ± 3.3 wt.% Sn, and 20.4 ± 8.1 wt.% Pb (18 measurements of the six coins). The heterogeneity of Group II's coins is presented in the ternary diagram of the alloy composition (Fig. 2b).

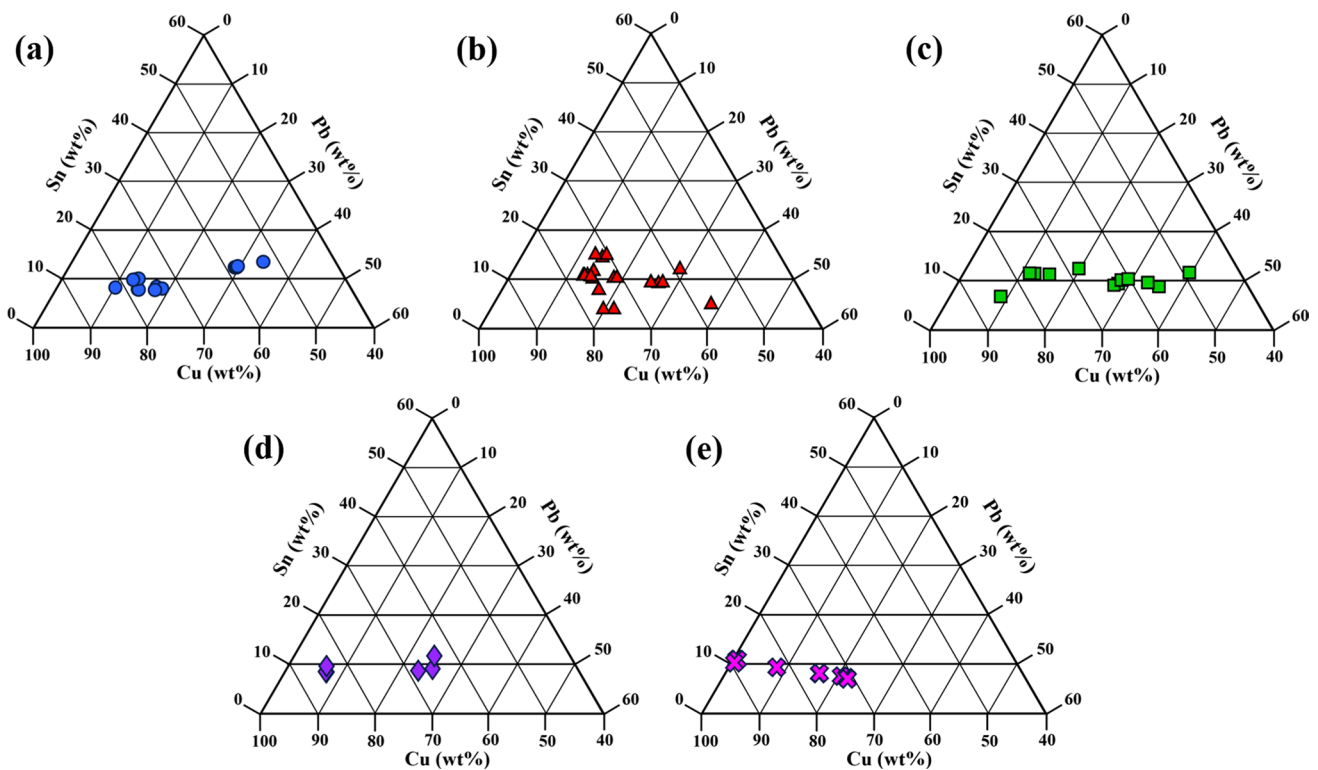
SEM observation of coin nos. 2.1, 2.2, and 2.3 exposed metal after local grinding of their surface revealed dark areas rich in copper and bright islands rich in lead according to BSE mode (Fig. 5). The spectrum of coin no. 2.4, area rich in copper according to BSE mode, revealed the presence of the elements Cu, Pb, Sn, C, and O (Fig. 6a), whereas no chlorine was detected in this examined area. SEM observation of coin nos. 2.4, 2.5, and 2.6 exposed ground metal also revealed dark areas rich in copper and bright islands rich in lead according to BSE mode. Marble-like texture was observed in some areas of coin 2.4 (Fig. 7a–b). In BSE mode, the ground (unetched) metallic surface of this coin exhibits a distinct dark and bright marbled texture (Fig. 7b). Given that BSE contrast is primarily controlled by the average atomic number, the very bright areas correspond to lead islands. The marbled texture observed in BSE mode can be explained by microsegregation during solidification of the Cu-Sn-Pb alloy. As the melt begins to solidify, the primary α -(Cu,Sn) phase initially crystallizes as dendrites that are relatively poor in tin and therefore appear darker in BSE images. As solidification proceeds, the remaining liquid becomes gradually enriched in tin and can result in coring of the α -phase. The remaining melt, now highly enriched in tin, forms the interdendritic phase ($\alpha + \delta$ eutectoid) that appears brighter in the BSE images. The two-phase character of Cu-Sn alloys is responsible for a heterogeneous microhardness that can be homogenized by annealing [56].

Table 2 XRF analysis results of the copper alloy coins from Tell Iztabba after sand polishing: Group I (coin nos. 1.1–1.4), Group II (coin nos. 2.1–2.6), Group III (coin nos. 3.1–3.4), Group IV (coin nos. 4.1, 4.2), Group V (coin nos. 5.1–5.3), and test samples (coin nos. T1–T3), where SA represents the scanned area

Sample description	Alloying elements (wt.%)			Impurities (wt.%)										
	Cu	Sn	Pb	Si	Ni	Fe	Co	Zn	As	Sb	W	Ti	Mg	Other
Coin no. 1.1, obverse, area 1	77.4	7.6	14.0	0.7	0.1	0.1	0.1	–	–	–	–	–	–	–
Coin no. 1.1, obverse, area 2	77.0	7.7	14.3	0.8	0.1	0.1	–	–	–	–	–	–	–	–
Coin no. 1.1, reverse	80.0	7.7	10.1	1.7	0.2	0.1	0.1	0.1	–	–	–	–	–	–
Coin no. 1.2, obverse, area 1	71.7	7.8	17.0	2.4	0.1	0.1	–	–	0.8	0.1	–	–	–	–
Coin no. 1.2, obverse, area 2	72.1	7.6	17.8	1.3	0.1	0.1	–	0.1	0.9	–	–	–	–	–
Coin no. 1.2, reverse	71.0	7.3	16.1	4.2	0.1	0.1	0.1	0.1	0.8	0.2	–	–	–	–
Coin no. 1.3, obverse, area 1	72.5	9.4	12.1	4.0	0.1	0.1	0.1	0.1	1.3	0.3	–	–	–	–
Coin no. 1.3, obverse, area 2	72.7	9.6	12.6	3.4	–	0.1	0.1	0.1	1.1	0.3	–	–	–	0.1 Ag
Coin no. 1.3, reverse	73.9	9.0	11.9	3.7	0.1	0.1	0.1	0.1	1.1	–	–	–	–	–
Coin no. 1.4, reverse, area 1	56.8	11.8	28.9	–	–	0.2	0.1	–	1.6	–	0.1	0.2	0.3	–
Coin no. 1.4, reverse, area 2	56.0	12.0	29.1	–	0.1	0.2	0.1	–	1.6	–	0.1	0.4	0.4	–
Coin no. 1.4, obverse	52.1	12.7	32.2	–	0.1	0.2	0.1	–	1.7	–	0.1	0.4	–	0.4 S
Coin no. 2.1, reverse, area 1	67.2	13.8	13.3	3.9	0.1	0.1	0.1	0.1	1.3	–	–	0.1	–	–
Coin no. 2.1, reverse, area 2	66.8	14.1	13.8	3.7	0.1	0.1	0.1	–	1.1	–	0.1	0.1	–	–
Coin no. 2.1, obverse	69.7	14.4	11.9	2.1	0.2	0.1	0.1	0.1	1.1	–	–	–	0.3	–
Coin no. 2.2, obverse, area 1	60.5	8.7	26.2	1.9	0.1	0.1	–	–	1.8	–	0.1	0.3	0.3	–
Coin no. 2.2, obverse, area 2	62.5	8.7	24.0	2.0	0.1	–	0.1	–	1.7	–	0.1	0.4	0.4	–
Coin no. 2.2, reverse	72.1	7.6	16.1	2.7	0.1	0.1	0.1	0.1	1.1	–	–	–	–	–
Coin no. 2.3, obverse, area 1	70.4	9.9	17.9	–	0.1	0.2	0.1	0.1	1.1	–	0.1	–	0.1	–
Coin no. 2.3, obverse, area 2	69.7	10.1	18.4	–	0.1	0.2	0.1	0.1	1.0	–	–	0.3	–	–
Coin no. 2.3, reverse	57.9	11.8	28.2	–	0.1	0.2	–	–	1.4	–	0.1	–	0.3	–
Coin no. 2.4, obverse, area 1	74.3	10.0	14.2	–	0.1	0.1	0.1	0.1	0.8	0.2	0.1	–	–	–
Coin no. 2.4, obverse, area 2	74.1	10.4	13.9	–	0.1	0.1	0.1	0.1	0.9	0.2	0.1	–	–	–
Coin no. 2.4, reverse	72.3	10.9	15.0	0.2	0.1	0.1	0.1	0.1	1.0	0.2	–	–	–	–
Coin no. 2.5, obverse, area 1	73.2	3.5	21.3	–	0.1	0.1	–	0.1	1.4	0.1	–	–	0.2	–
Coin no. 2.5, obverse, area 2	75.7	3.5	19.3	–	0.2	0.1	–	0.1	1.0	0.1	–	–	–	–
Coin no. 2.5, reverse	55.7	4.0	37.6	–	–	0.1	–	–	1.9	–	0.1	0.6	–	–
Coin no. 2.6, reverse, area 1	72.8	10.0	12.8	2.7	0.2	0.3	0.1	0.1	1.0	–	–	–	–	–
Coin no. 2.6, reverse, area 2	72.9	9.7	13.4	2.5	0.1	0.2	0.1	0.1	1.0	–	–	–	–	–
Coin no. 2.6, obverse	49.5	7.9	37.5	2.1	–	0.2	–	–	1.7	–	0.1	0.7	0.2	0.1 Zr
Coin no. 3.1, obverse, area 1	60.9	8.4	27.8	–	–	0.1	0.1	–	1.9	0.2	0.1	0.4	–	0.1 Ag
Coin no. 3.1, obverse, area 2	62.2	8.1	27.1	–	–	0.1	0.1	–	1.7	0.2	–	–	0.3	0.2 Ag
Coin no. 3.1, reverse	82.8	6.6	9.2	–	–	0.1	–	0.1	1.0	0.2	–	–	–	–
Coin no. 3.2, obverse, area 1	59.0	9.2	27.1	2.4	–	0.1	0.1	–	1.7	–	0.1	0.3	–	–
Coin no. 3.2, obverse, area 2	57.3	9.9	28.0	2.4	–	0.1	0.1	–	1.7	–	0.1	0.4	–	–
Coin no. 3.2, reverse	69.6	10.4	14.5	3.1	–	0.2	0.2	0.1	1.4	0.2	–	–	0.3	–
Coin no. 3.3, obverse, area 1	74.1	11.0	12.0	1.5	–	0.1	0.1	0.1	1.0	–	0.1	–	–	–
Coin no. 3.3, obverse, area 2	73.9	10.8	11.3	2.4	0.1	0.2	0.1	0.1	1.0	–	–	–	–	0.1 Ag
Coin no. 3.3, reverse	64.8	12.0	18.5	2.4	0.1	0.2	0.1	0.1	1.5	–	0.1	0.2	–	–
Coin no. 3.4, obverse, area 1	53.9	8.3	34.7	–	–	0.1	–	–	1.6	–	0.1	0.7	0.5	0.1 Ag
Coin no. 3.4, obverse, area 2	55.9	8.8	32.5	–	–	0.1	0.1	–	1.4	–	0.1	0.5	0.5	0.1 Ag
Coin no. 3.4, reverse	47.2	11.1	38.5	–	–	–	0.1	–	1.8	–	0.1	0.6	0.3	0.1 Cd, 0.2 S
Coin no. 4.1, reverse, area 1	64.9	8.0	22.8	2.4	0.2	0.1	0.2	–	1.1	–	0.1	0.2	–	–
Coin no. 4.1, reverse, area 2	63.5	8.5	25.1	0.6	–	0.1	0.2	–	1.3	–	0.1	0.3	0.3	–
Coin no. 4.1, obverse	61.5	11.1	23.8	0.3	–	0.1	0.1	–	2.0	–	0.1	0.4	0.6	–

Table 2 (continued)

Sample description	Alloying elements (wt.%)			Impurities (wt.%)										
	Cu	Sn	Pb	Si	Ni	Fe	Co	Zn	As	Sb	W	Ti	Mg	Other
Coin no. 4.2, obverse, area 1	82.1	7.8	7.1	2.3	–	0.1	–	0.1	0.5	–	–	–	–	–
Coin no. 4.2, obverse, area 2	82.2	7.9	7.0	1.9	–	0.2	0.1	0.1	0.4	–	–	–	–	0.1 Ag, 0.1 Cd
Coin no. 4.2, reverse	78.5	8.6	6.4	5.2	–	0.2	0.1	0.1	0.9	–	–	–	–	–
Coin no. 5.1, reverse, area 1	65.5	6.6	18.6	6.6	–	–	0.1	0.1	1.7	0.3	0.1	0.3	–	0.1 Ag
Coin no. 5.1, reverse, area 2	65.1	6.4	19.7	6.2	–	0.1	0.1	0.1	1.8	0.3	0.1	–	–	0.1 Ag
Coin no. 5.1, obverse	76.5	6.3	11.5	3.6	–	0.1	0.1	0.1	1.3	0.3	–	–	–	0.1 Ag, 0.1 Cd
Coin no. 5.2, reverse, area 1	86.7	10.4	0.7	1.0	0.1	0.1	–	0.1	0.8	–	–	–	–	0.1 Ag
Coin no. 5.2, reverse, area 2	86.9	10.4	0.8	0.8	0.1	0.1	–	0.1	0.8	–	–	–	–	–
Coin no. 5.2, obverse	87.9	9.8	0.7	0.5	0.1	0.1	–	0.1	0.7	–	–	–	–	0.1 Ag
Coin no. 5.3, reverse, area 1	66.7	6.8	14.8	10.1	–	0.5	0.2	0.1	0.7	–	0.1	–	–	–
Coin no. 5.3, reverse, area 2	66.7	6.9	14.6	10.4	–	0.5	0.2	0.1	0.6	–	–	–	–	–
Coin no. 5.3, obverse	77.8	8.4	8.2	4.0	–	0.3	0.3	0.1	0.9	–	–	–	–	–


Fig. 2 Ternary composition (wt.%) diagram of Cu-Sn-Pb of the coins from Tell Izṭabba based on XRF analysis: (a) Group I; (b) Group II; (c) Group III; (d) Group IV; and (e) Group V

The lead, which is essentially immiscible in copper, segregates into discrete precipitations during the final stages of solidification and appears as very bright islands due to its high atomic number. This order of solidification results in

compositional heterogeneity within the coin, resulting in the marbled contrast observed in the BSE micrographs (Fig. 7b).

Areas 1 and 2 (Fig. 7c–d) and area 3 (Fig. 7e) of coin no. 2.5, and areas 1–3 of coin 2.6 (Fig. 7f) were examined by

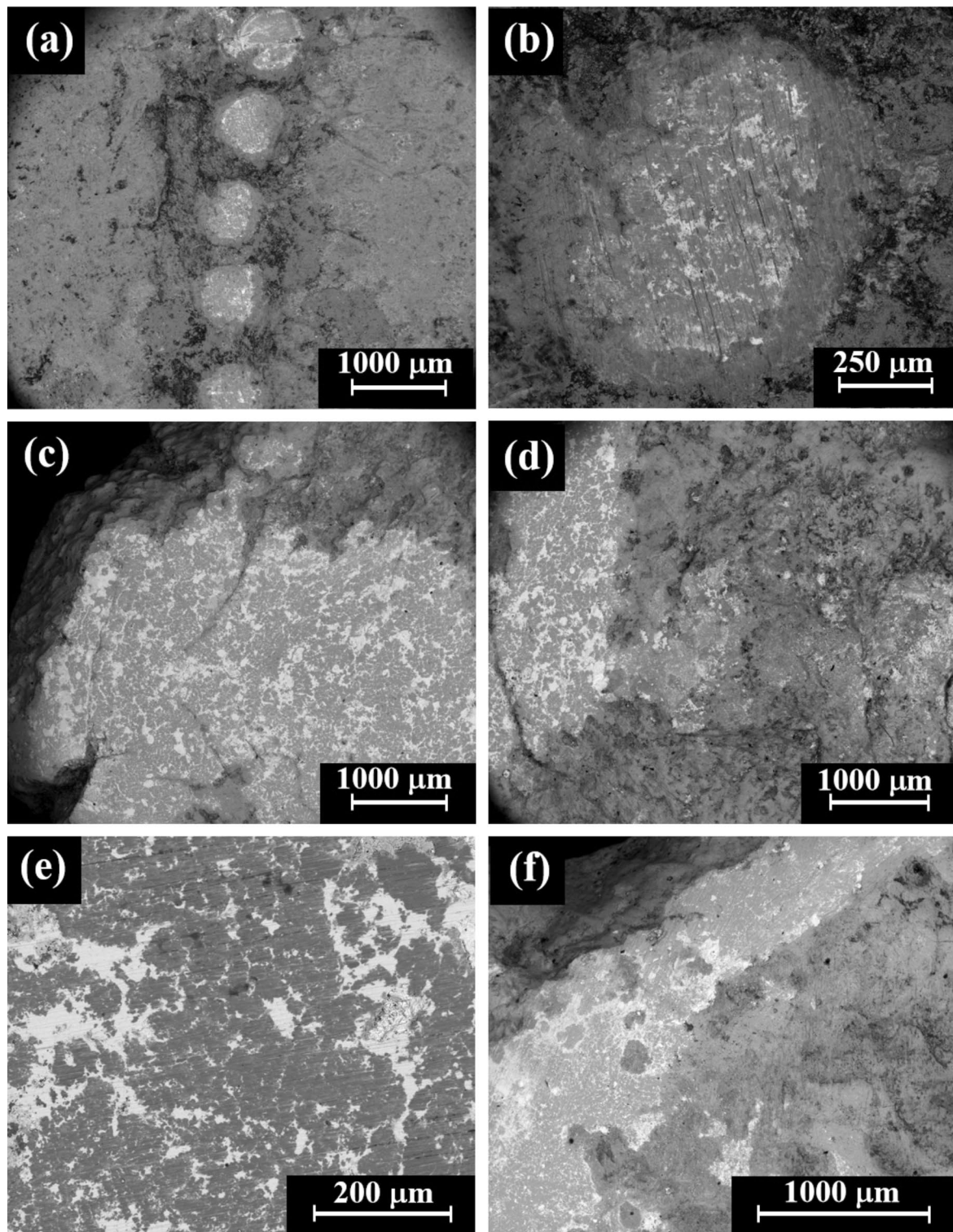


Fig. 3 SEM (BSE mode) images of the obverse of Group I's samples: (a) coin no. 1.1's surface, left side of the coin, where the areas of the bright circles were ground; (b) higher magnification of the surface of coin no. 1.1, showing one of the ground circles; (c) coin no. 1.2's surface, lower part of the coin after grinding (examined area 1);

(d) coin no. 1.2's surface, upper part of the coin, near the circumference (examined area 2), showing the ground area at left side of the image; (e) higher magnification of coin no. 1.2's surface (area 2); and (f) coin no. 1.3's surface, upper part of the coin after grinding (bright area at the left side of image)

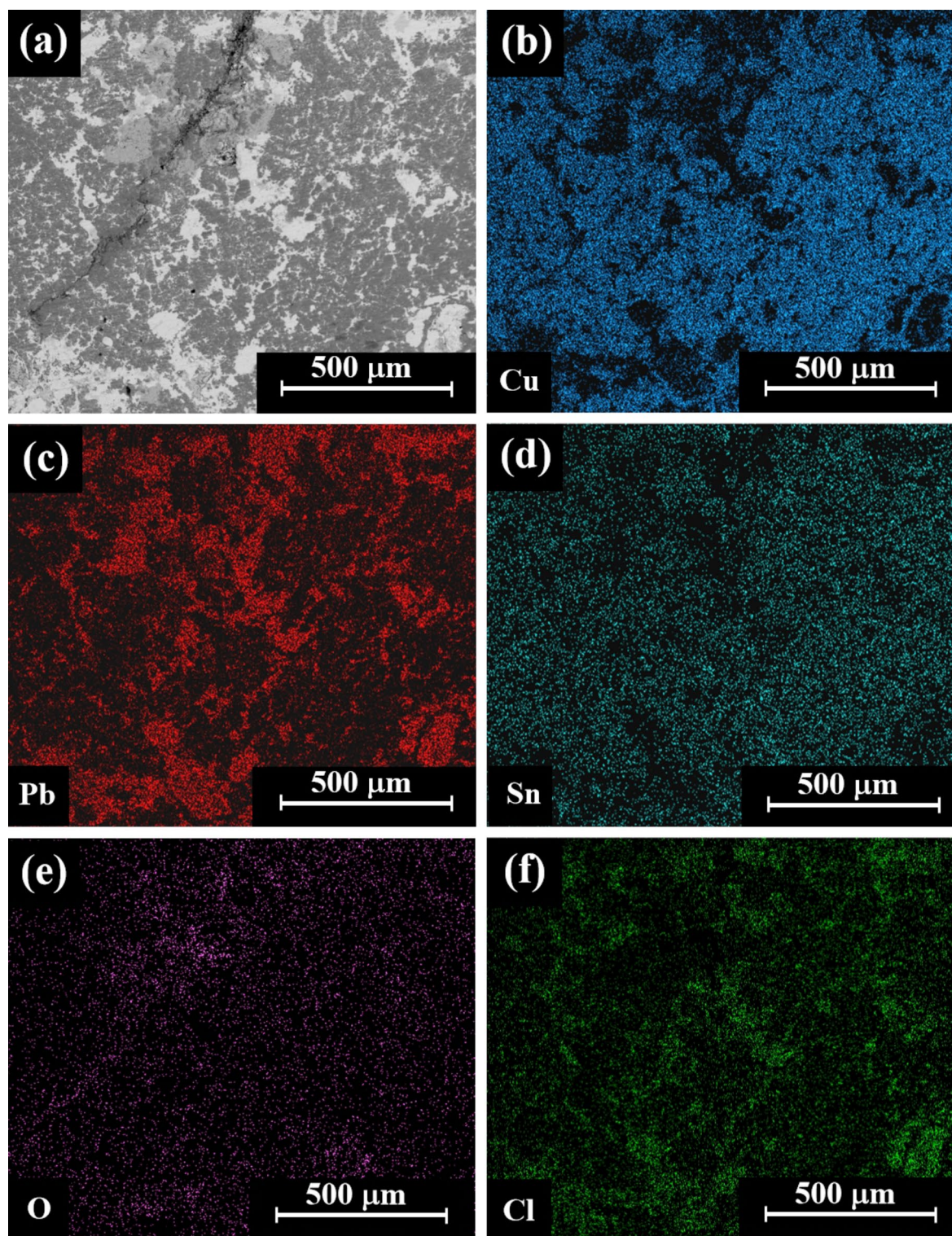


Fig. 4 SEM-EDS elemental mapping of the lower part of the coin no. 1.2's surface (obverse after grinding, area 1), showing the presence of: (a) dark areas of copper and bright areas of lead (according to

BSE mode), and the presence of the following elements (bright dots): (b) copper, (c) lead, (d) tin, (e) oxygen, and (f) chlorine

Table 3 SEM-EDS analysis results of the ground areas of the copper alloy coins from Tell Izṭabba: Group I (coin nos. 1.1–1.4), Group II (coin nos. 2.1–2.6), Group III (coin nos. 3.1–3.4), Group IV (coin

nos. 4.1, 4.2), Group V (coin nos. 5.1–5.3), and test samples (coin nos. T1–T3), where SA represents the scanned area

Sample No. and description of the examined area	Alloy composition (wt.%)			
	Cu	Sn	Pb	Other elements
Coin no. 1.1, obverse, left side of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	69.0	12.9	18.1	–
Coin no.1.1, obverse, left side of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	73.3	17.2	9.5	–
Coin no.1.1, obverse, left side of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	72.0	12.5	15.5	–
Coin no.1.2, obverse, lower part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	51.8	5.9	42.3	–
Coin no.1.2, obverse, upper part of the coin near the circumference, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	60.5	6.3	33.2	–
Coin no.1.2, obverse, upper part of the coin near the circumference, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	56.1	7.0	36.9	–
Coin no.1.3, obverse, upper left part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$)	59.5	16.2	24.3	–
Coin no. 1.3, obverse, upper left part of the coin, SA 2: 300 $\mu\text{m} \times 300 \mu\text{m}$)	72.2	9.4	18.3	–
Coin no. 1.3, obverse, upper left part of the coin, SA 3: 300 $\mu\text{m} \times 300 \mu\text{m}$)	72.4	8.9	18.7	–
Coin no. 1.4, reverse, right side of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	56.7	9.8	33.5	–
Coin no. 1.4, reverse, right side of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	72.8	9.1	18.1	–
Coin no. 1.4, reverse, right side of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	51.3	10.3	38.4	–
Coin no. 2.1, reverse, lower right side of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	61.6	19.8	18.6	–
Coin no. 2.1, reverse, lower right side of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	63.6	17.4	19.0	–
Coin no. 2.1, reverse, lower right side of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	65.0	12.2	22.8	–
Coin no. 2.2, obverse, lower right part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	19.3	9.7	71.0	–
Coin no. 2.2, obverse, lower right part of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	41.0	27.5	31.5	–
Coin no. 2.2, obverse, lower right part of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	73.5	7.0	19.5	–
Coin no. 2.3, observe, lower right part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	65.3	9.5	25.2	–
Coin no. 2.3, observe, upper right part of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	68.4	14.3	17.4	–
Coin no. 2.3, observe, upper right part of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	70.2	17.3	12.5	–
Coin no. 2.4, observe, center of coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	76.3	15.3	8.4	–
Coin no. 2.4, observe, right side of the coin near its circumference, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	80.5	13.0	6.5	–
Coin no. 2.4, observe, right side of the coin near its circumference, marble-like texture, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	54.2	8.6	37.2	–
Coin no.2.4, observe, dark area, left side of the coin, SA 4: 500 $\mu\text{m} \times 500 \mu\text{m}$	89.4	7.4	3.2	–
Coin no. 2.5, observe, center of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	84.6	3.1	12.2	–
Coin no. 2.5, observe, center of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	69.4	4.7	25.9	–
Coin no. 2.5, observe, right side of the coin near its circumference, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	84.0	4.0	12.0	–
Coin no. 2.6, reverse, upper part of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	68.3	9.4	22.3	–
Coin no. 2.6, reverse, upper part of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	55.6	24.2	20.5	–
Coin no. 2.6, reverse, upper part of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	75.5	8.4	16.1	–
Coin no. 3.1, observe, upper left part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	74.4	5.6	19.8	–
Coin no. 3.1, observe, left part of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	77.0	6.2	16.8	–
Coin no. 3.1, observe, left part of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	41.0	5.9	53.1	–
Coin no. 3.1, observe, upper right part of the coin, SA 4: 500 $\mu\text{m} \times 500 \mu\text{m}$	84.1	7.3	8.6	–
Coin no. 3.1, observe, dark area at the upper right part of the coin, SA 5: 500 $\mu\text{m} \times 500 \mu\text{m}$	61.8	2.3	2.7	33.2 Cl
Coin no. 3.2, observe, upper part of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	58.2	11.6	30.2	–
Coin no. 3.2, observe, upper part of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	67.4	8.1	24.5	–
Coin no. 3.2, observe, left part of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	70.5	8.1	21.4	–
Coin no. 3.3, observe, left part of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	71.1	5.6	23.3	–
Coin no. 3.3, observe, left part of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	66.8	6.1	27.0	–
Coin no. 3.3, observe, left part of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	61.0	5.1	33.9	–
Coin no. 3.3, observe, fractured area at the center of the coin, SA 4: 30 $\mu\text{m} \times 30 \mu\text{m}$	54.7	4.2	7.9	23.6 C, 6.9 O, 2.7 Cl
Coin no. 3.4, observe, upper left part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	70.1	7.8	22.1	–

Table 3 (continued)

Sample No. and description of the examined area	Alloy composition (wt.%)			
	Cu	Sn	Pb	Other elements
Coin no. 3.4, observe, upper left part of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	67.0	8.6	24.4	–
Coin no. 3.4, observe, upper left part of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	60.8	10.7	28.6	–
Coin no. 4.1, reverse, upper part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	70.1	8.3	21.6	–
Coin no. 4.1, reverse, upper part of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	64.5	8.5	27.0	–
Coin no. 4.1, reverse, inside the letter A, SA 3: 450 $\mu\text{m} \times 450 \mu\text{m}$	61.1	10.2	28.7	–
Coin no. 4.2, observe, lower right part of coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	75.9	6.9	17.2	–
Coin no. 4.2, observe, right side of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	81.0	8.0	11.0	–
Coin no. 4.2, observe, upper right part of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	80.6	7.9	11.5	–
Coin no. 5.1, reverse, lower part of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	73.2	5.7	21.1	–
Coin no. 5.1, reverse, lower part of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	72.2	5.6	22.2	–
Coin no. 5.1, reverse, lower part of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	69.0	5.7	25.3	–
Coin no. 5.2, reverse, center of the coin, SA 1: 500 $\mu\text{m} \times 500 \mu\text{m}$	89.6	10.4	–	–
Coin no. 5.2, reverse, left side of the coin, SA 2: 500 $\mu\text{m} \times 500 \mu\text{m}$	89.9	10.1	–	–
Coin no. 5.2, reverse, right side of the coin, SA 3: 500 $\mu\text{m} \times 500 \mu\text{m}$	89.8	10.2	–	–
Coin no. 5.3, reverse, left side of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	68.2	7.0	24.8	–
Coin no. 5.3, reverse, center side of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	63.1	7.4	29.5	–
Coin no. 5.3, reverse, right side of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	85.5	8.6	5.9	–
Coin no. T1, exposed metal, upper left side of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	70.1	7.1	22.8	–
Coin no. T1, upper left side of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	62.7	7.5	29.8	–
Coin no. T1, upper left side of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	59.3	7.0	33.7	–
Coin no. T2, left side of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	89.3	4.2	6.5	–
Coin no. T2, left side of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	52.0	5.2	42.8	–
Coin no. T2, left side of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	58.9	6.1	35.0	–
Coin no. T3, left side of the coin, SA 1: 400 $\mu\text{m} \times 400 \mu\text{m}$	81.5	13.3	5.2	–
Coin no. T3, left side of the coin, SA 2: 400 $\mu\text{m} \times 400 \mu\text{m}$	85.8	11.5	2.7	–
Coin no. T3, left side of the coin, SA 3: 400 $\mu\text{m} \times 400 \mu\text{m}$	81.8	13.3	4.9	–

Table 4 ICP-MS analysis results of the dissolved powder drilled from the coins' bulk: Coin nos. 1.2 (Group I), 2.6 (Group II), 3.2 (Group III), 4.1 (Group IV), 5.1 and 5.2 (Group V)

Sample	Alloying elements (wt. %)			Impurities (wt.%)												
	Cu	Sn	Pb	Ag	Sb	Te	Hg	Bi	Au	P	S	Ni	Co	Zn	As	Se
Coin no. 1.2	72.2	5.88	21.2	0.045	0.109	0.002	0.00009	0.007	0.003	0.000	0.105	0.126	0.061	0.003	0.252	0.001
Coin no. 2.6	72.4	6.63	20.2	0.041	0.035	0.003	0.00012	0.002	0.004	0.002	0.187	0.161	0.091	0.009	0.285	0.002
Coin no. 3.2	73.6	7.19	18.6	0.047	0.098	0.003	0.00010	0.008	0.004	0.001	0.167	0.052	0.057	0.007	0.285	0.000
Coin no. 4.1	71.0	6.31	22.0	0.029	0.052	0.002	0.00010	0.003	0.003	0.001	0.172	0.051	0.113	0.004	0.263	0.003
Coin no. 5.1	69.9	4.29	24.8	0.052	0.252	0.002	0.00009	0.023	0.003	0.002	0.080	0.054	0.044	0.005	0.434	0.005
Coin no. 5.2	90.1	8.61	0.30	0.043	0.068	0.007	<0.00006	0.002	0.005	0.006	0.029	0.170	0.010	0.002	0.691	0.004

EDS analysis. EDS results after sandblasting followed by locally grinding the surface revealed a compositional heterogeneity (Table 3), where the average composition of Group II's coins is 66.6 ± 15.9 wt.% Cu, 12.2 ± 6.5 wt.% Sn, and 21.1 ± 14.3 wt.% Pb (19 measurements of ground areas).

ICP-MS analysis of Coin no. 2.6 detected that its bulk composition is composed of Cu-Sn-Pb alloy, consisting of 72.4 wt.% Cu, 6.6 wt.% Sn, and 20.2 wt.% Pb as the major alloying elements. Minor and trace elements were also identified in this coin, including minor quantities of S, Ni, and As, and traces of the elements Ag, Sb, Te, Hg, Bi, Au, P, Co, Zn, and Se (Table 4).

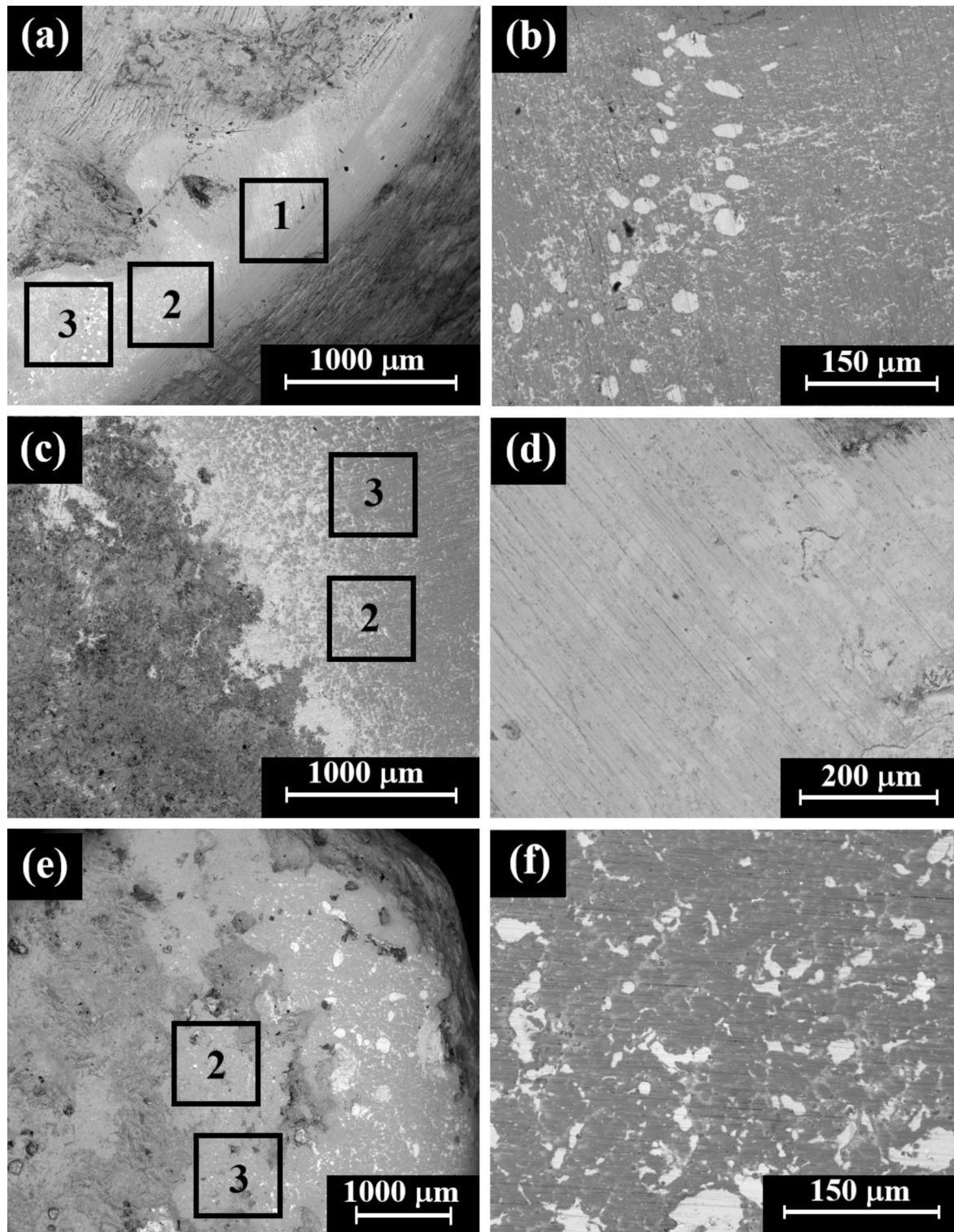


Fig. 5 SEM images of Group II's samples: **(a)** coin no. 2.1's surface, lower right part of the coin (reverse, BSE mode), where the ground areas 1–3 were examined by EDS analysis; **(b)** coin no. 2.1's ground surface, higher magnification of area 3 in Fig. 4a (BSE mode); **(c)** coin no. 2.2's surface, lower right part of the coin (observe, examined areas 2–3, BSE mode), where the right side of the image was

ground; **(d)** coin no. 2.2's ground surface, upper left part of the coin (observe, examined area 1, SE mode); **(e)** coin no. 2.3's ground surface, upper right part of the coin (observe, examined areas 2–3, BSE mode); and **(f)** coin no. 2.3's ground surface, lower right part of the coin (observe, examined area 1, BSE mode)

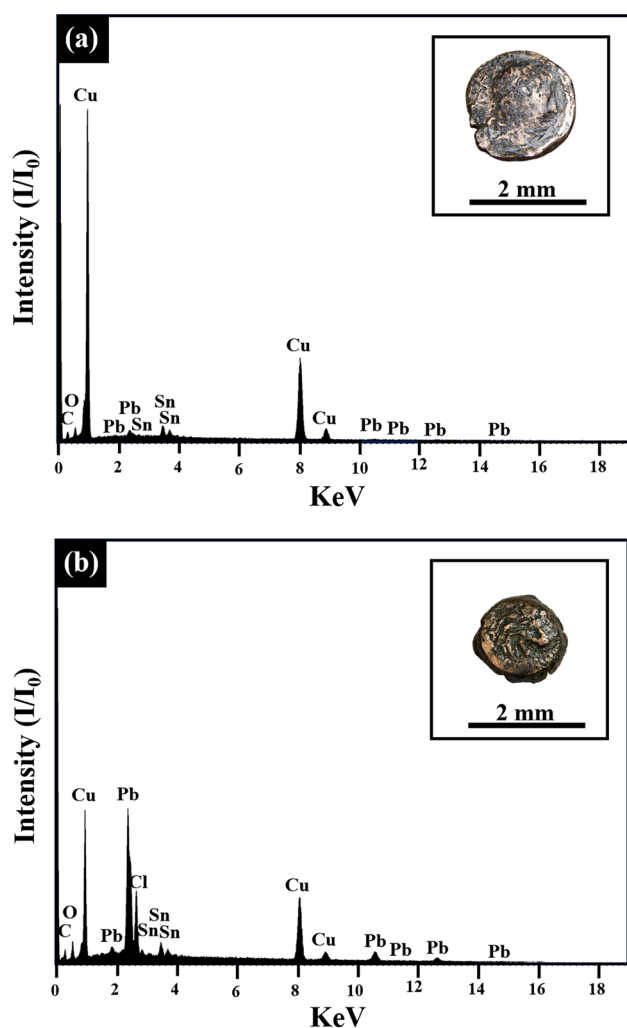


Fig. 6 SEM-EDS chemical analysis, showing the spectrum of two typical copper alloy coins from Tell Iztabba: **(a)** Group II, coin no. 2.4, examined area rich with copper; and **(b)** Group III, coin no. 3.1, examined rich with lead and chlorine

Coin Nos. 3.1, 3.2, 3.3, 3.4 (Group III = SC 2067, 2068, Antiochos VII, mint of Antioch)

Group III of the mint of Antioch on the Orontes includes four copper alloy coins (Fig. 1). The coins' weight ranges between 2.80 and 5.83 g. The description of the images on the obverse and reverse of Group III's coins is presented in Table 1.

The XRF results after sandblasting indicate the heterogeneity of Group III's coins (Table 2), where the average composition of this group is 65.8 ± 9.8 wt.% Cu, 9.9 ± 1.6 wt.% Sn, and 24.3 ± 9.8 wt.% Pb (15 measurements of the four coins). The compositional heterogeneity of Group III's coins is shown in the ternary composition diagram (Fig. 2c).

SEM observation of Group III's coins revealed dark areas rich in copper and bright areas rich in lead according to BSE

mode (Figs. 8 and 9). Area 1 (Fig. 8a–b) and areas 2 and 3 (Fig. 8c) of coin no. 3.1, and areas 1–3 (Fig. 8d–f) of coin no. 3.2 were examined by EDS analysis. Areas 1–3 of coin no. 3.3 (Fig. 9a–b) and area 1 of coin no. 3.4 (Fig. 9c–d) were also examined by EDS analysis.

The spectrum of coin no. 3.1, area rich in lead according to BSE mode, revealed the presence of the elements Cu, Pb, Sn, Cl, C, and O (Fig. 6b), showing there is a correlation between the presence of lead and chlorine. EDS analysis results of Group III's coins alloy composition after sandblasting followed by locally grinding the surface indicate on the heterogeneity of this group of coins (Table 3). The average composition of Group III's coins according to EDS analysis is 66.9 ± 10.1 wt.% Cu, 7.4 ± 1.9 wt.% Sn, and 25.7 ± 10.0 wt.% Pb (13 measurements of ground areas). Two areas rich in chlorine were not included in the average composition calculations (Table 3).

ICP-MS analysis of Coin no. 3.2 exposed that its base metal is made of Cu-Sn-Pb alloy, consisting of 73.6 wt.% Cu, 7.2 wt.% Sn, and 18.6 wt.% Pb as the main alloying elements. Minor and trace elements were also discovered in this coin, including minor amount of S and As, and traces of the elements Ag, Sb, Te, Hg, Bi, Au, P, Ni, Co, Zn, and Se (Table 4).

Coin Nos. 4.1, 4.2 (Group IV = SC 2300, 2307, Antiochos VIII, mint of Antioch)

Group VI of the mint of Antioch on the Orontes includes two copper alloy coins (Fig. 1). The coins' weight ranges between 4.33 and 6.11 g. The description of the images on the obverse and reverse of Group IV's coins is given in Table 1.

XRF results of Group IV's coins after sandblasting indicate the heterogeneity of this group of coins (Table 2). The average composition of Group IV's coins according to XRF analysis is 75.0 ± 9.4 wt.% Cu, 9.0 ± 1.2 wt.% Sn, and 16.0 ± 8.9 wt.% Pb (6 measurements of ground areas of two coins). The compositional heterogeneity of the measured values of Group IV's coins is presented in the ternary composition diagram (Fig. 2d).

The Greek letters of the word *ANTIOX* on the surface of coin no. 4.1 are brighter than the oxide that surrounds them (Fig. 10a–c). SEM observation of Group IV's coins revealed dark areas rich in copper and bright islands rich in lead according to BSE mode (Fig. 10d, coin no. 4.1). In BSE imaging, brighter regions correspond to elements with higher average atomic number, while darker regions indicate elements with lower atomic number; thus, the contrast directly reflects the relative lead and copper content of the coin's surface.

EDS analysis results of Group IV's coins alloy composition after sandblasting followed by locally grinding the

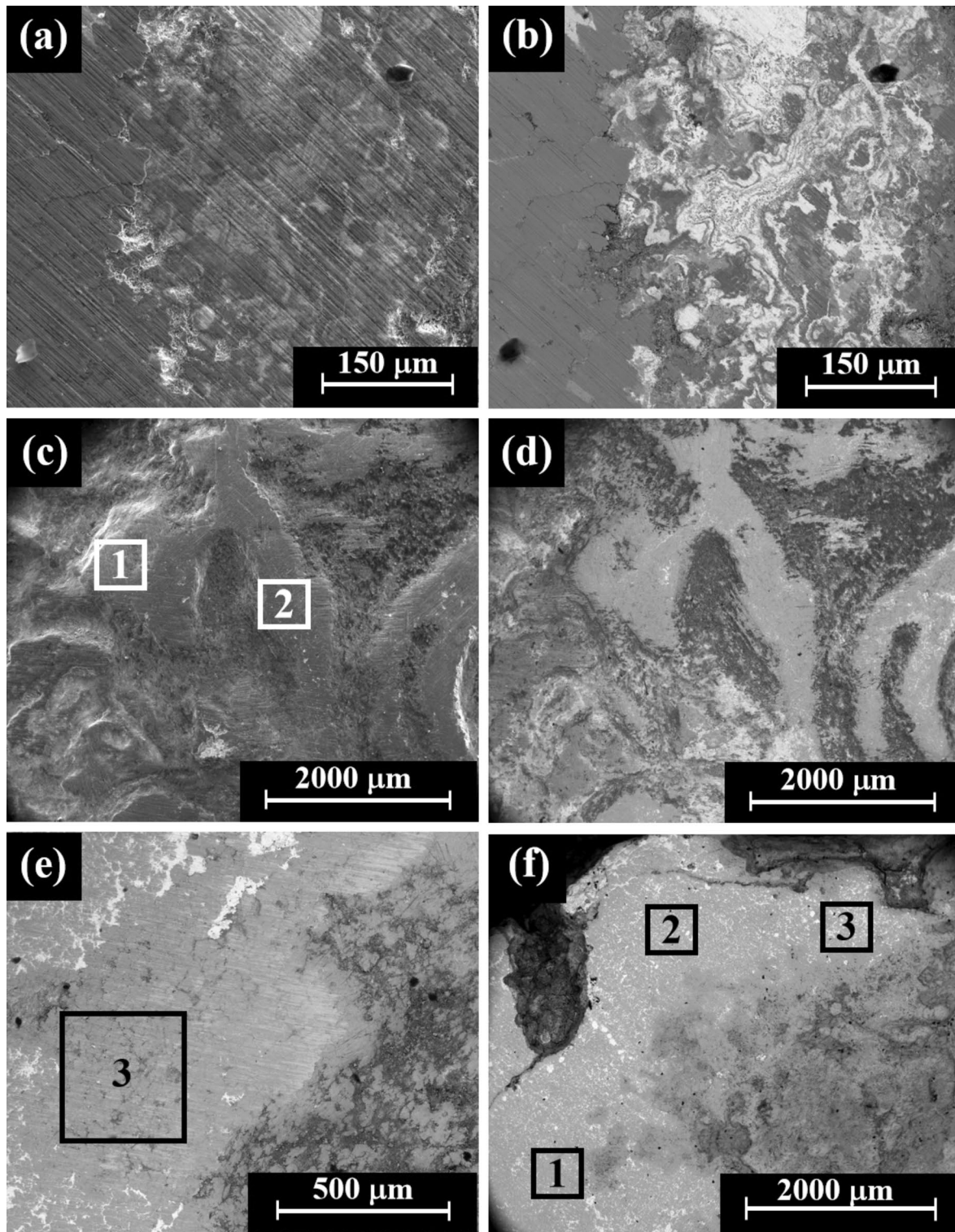


Fig. 7 SEM images of Group II's samples: **(a, b)** coin no. 2.4's ground surface, right side of the coin's obverse, examined area of marble-like texture (SE and BSE mode, respectively); **(c, d)** general view of the centre of coin no. 2.5's obverse surface (SE and BSE mode, respectively), where areas 1 and 2 were examined by EDS; **(e)**

coin no. 2.5's obverse surface, right side of the coin, near its circumference (BSE mode, examined area 3); and **(f)** coin no. 2.6's surface, upper part of the coin's reverse, where areas 1–3 were examined by EDS

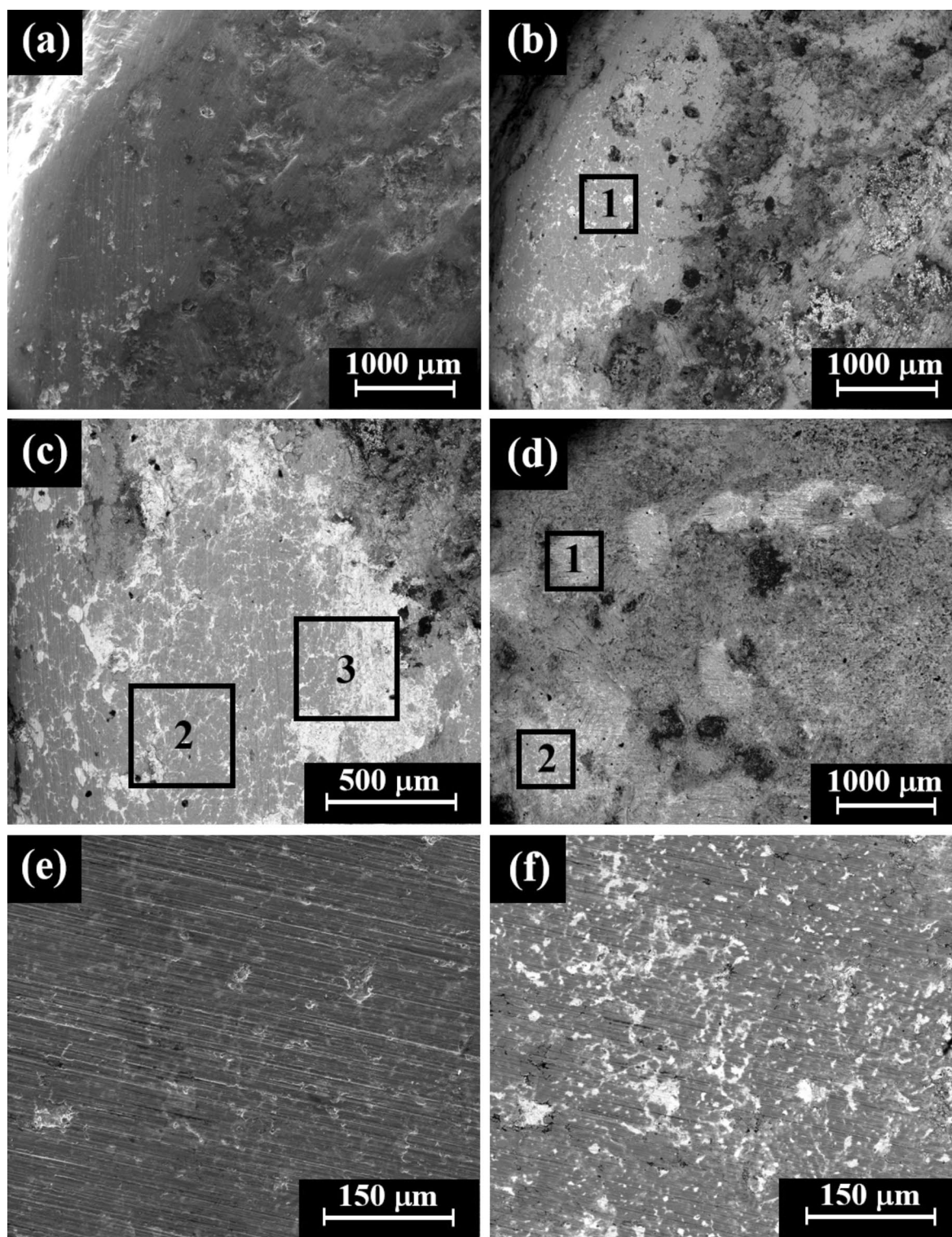


Fig. 8 SEM images of Group III's samples (observe): (a, b) coin no. 3.1's ground surface, upper left part of the images, examined area 1 (SE and BSE mode, respectively); (c) coin no. 3.1's ground surface, left part of the coin, examined areas 2 and 3; (d) coin no. 3.2's

ground surface, upper part of the coin, examined areas 1 and 2; and (e, f) coin no. 3.2's ground surface, left part of the coin, examined area 3

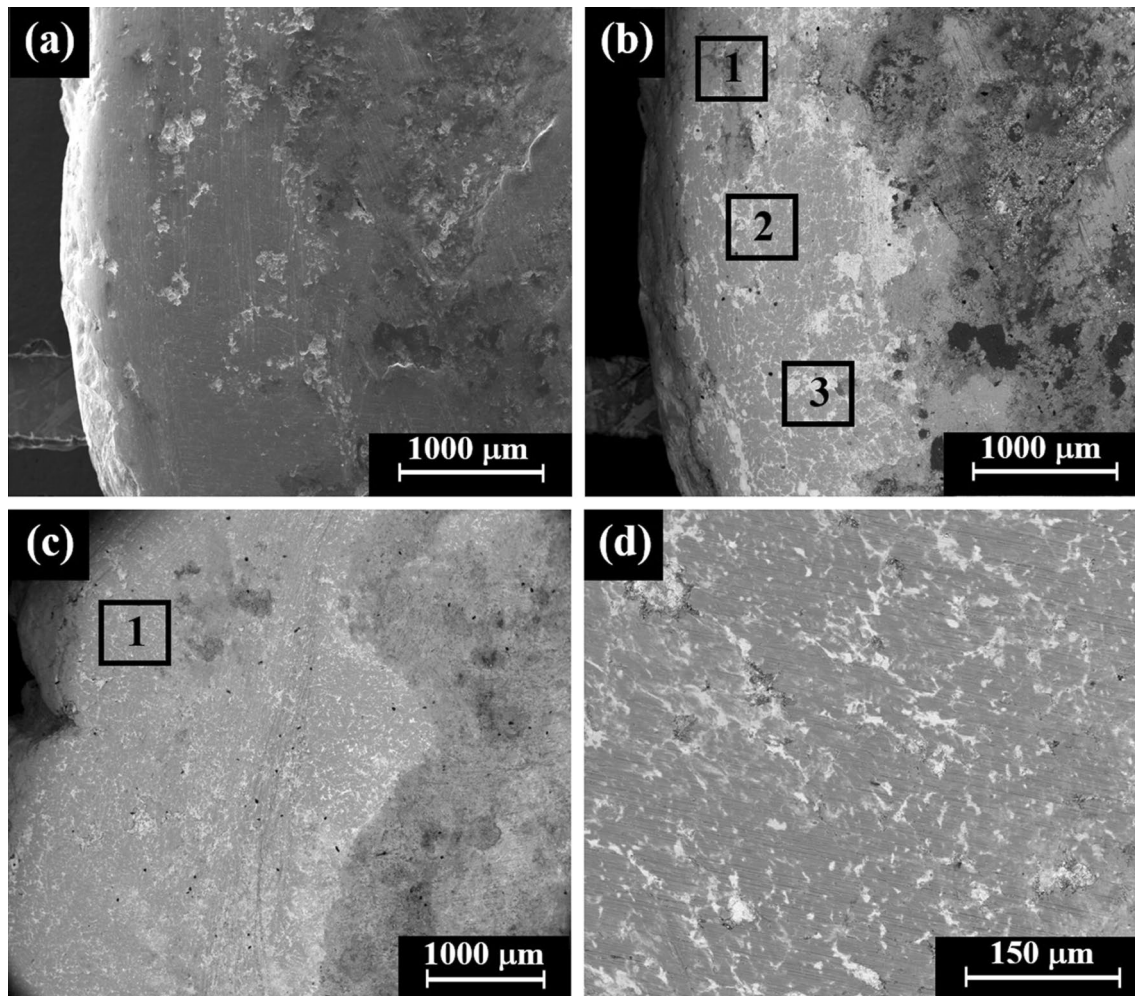


Fig. 9 SEM images of Group III's samples: (a, b) coin no. 3.3, reverse, ground surface, left side of the coin, examined area 1–3; (c) general view of coin no. 3.4, observe, ground surface, left side part

of the coin, showing examined area 1; and (d) coin no. 3.4, observe, higher magnification of examined area 1

surface indicate the heterogeneity of Group IV's coins (Table 3). The average composition of Group IV's coins according to EDS analysis is 72.2 ± 7.6 wt.% Cu, 8.3 ± 1.0 wt.% Sn, and 19.5 ± 6.9 wt.% Pb (six measurements of ground areas).

ICP-MS analysis of Coin no. 4.1 exposed that its base alloy is made of Cu-Sn-Pb, consisting of 71.0 wt.% Cu, 6.3 wt.% Sn, and 22.0 wt.% Pb as the major alloying elements. Minor amount of S, Co, and As, and traces of the elements Ag, Sb, Te, Hg, Bi, Au, P, Ni, Zn, and Se were also detected (Table 4).

Coin Nos. 5.1, 5.2, 5.3 (Group V = SC 1672, 1676 [Nos. 5.2, 5.3]; Demetrius I, mint of Tyre and SC 2119 [No. 5.1]; Antiochos VII, mint of Aké-Ptolemaïs)

Group V (Aké-Ptolemaïs mint for Coin no. 5.1 and mint of Tyre for coin nos. 5.2 and 5.3) includes three copper-based

coins (Fig. 1). The coins' weight ranges between 2.97 g and 7.04 g. The description of the images on the obverse and reverse of Group V's coins is given in Table 1.

XRF analysis results of Group V's coin nos. 5.1 and 5.3 after sandblasting revealed they were made of ternary copper-tin-lead alloy (Table 2). The XRF analysis results on these two coins revealed their heterogeneity. The average composition of Group V's coins 5.1 and 5.3 according to XRF analysis is 76.4 ± 4.1 wt.% Cu, 7.6 ± 0.7 wt.% Sn, and 16.1 ± 4.5 wt.% Pb (6 measurements of the two coins). XRF analysis results of coin 5.2 after sandblasting indicate that it is composed of a binary copper-tin alloy, containing only a trace amount of lead (less than 1 wt.%) (Table 2). According to XRF analysis, the average composition of coin no. 5.2 is 88.9 wt.% Cu, 10.4 wt.% Sn, and 0.7 wt.% Pb (three measurements of this coin were done). The compositional heterogeneity of Group V coins is illustrated in the ternary alloy diagram, showing the variation in Cu, Sn,

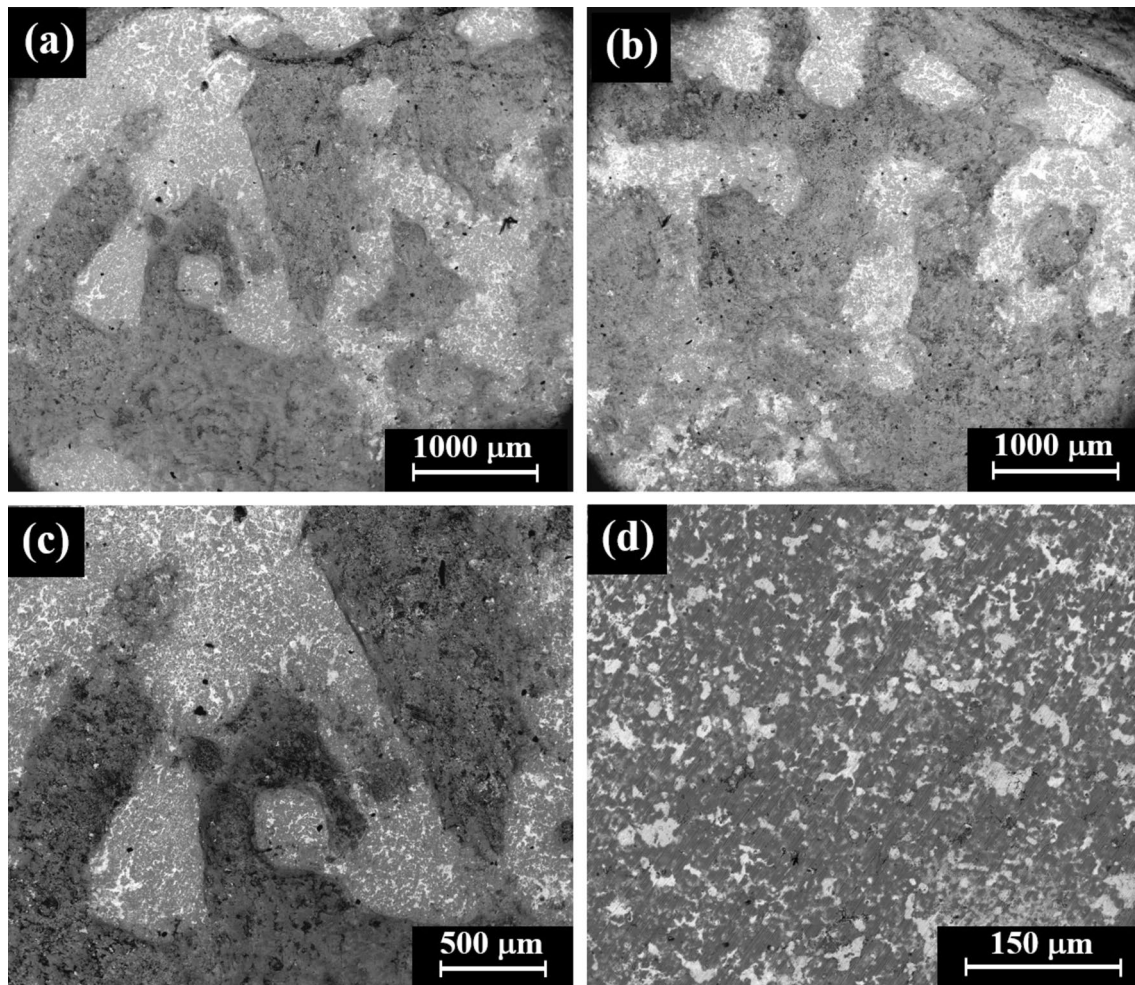


Fig. 10 SEM BSE mode images of coin no. 4.1 (Group IV), reverse side after grinding the surface: (a, b) the letters *ANTIOX*; (c) higher magnification of the letter A; and (d) examined area 3 inside the letter A

and Pb concentrations across the group, ranging from the binary Cu-Sn composition of coin 5.2 to the ternary Cu-Sn-Pb alloys of coins 5.1 and 5.3 (Fig. 2e).

SEM observation of Group V's coin no. 5.1 (Fig. 11a, areas 1–3 examined by EDS analysis) revealed dark areas rich in copper and brightislands rich in lead according to BSE mode (Fig. 11b). The struck Greek letters of the words *ΒΑΣΙΛΕΥΣ* and *ΔΗΜΗΤΡΙΟΥ* were observed in coin 5.2 (Fig. 11c–d, BSE mode). SEM observation of coin 5.3 reverse shows branches of a palm tree (Fig. 12a) and the letter “N” at the lower part of the coin (Fig. 12b), where NP represents the date 154 BCE. SEM observation of coin 5.3's ground surface near the circumference (area 1, Fig. 12c) revealed dark areas rich in copper and brightislands rich in lead according to BSE mode (Fig. 12d). In BSE images, brighter areas indicate heavier elements, while darker areas indicate lighter elements, reflecting the distribution of lead and copper on the coin's surface.

EDS results of Group V's coin nos. 5.1 and 5.3 alloy composition after sandblasting followed by locally grinding the surface revealed the heterogeneity of these two coins that are made of ternary copper-tin-lead alloy (Table 3). The average composition of coins 5.1 and 5.3 according to EDS analysis is 71.8 ± 6.9 wt.% Cu, 6.7 ± 1.1 wt.% Sn, and 21.5 ± 7.5 wt.% Pb (six measurements of ground areas). According to EDS analysis, Group V's coin no. 5.2 was made of binary copper-tin (Table 2), where the average composition of coin no. 5.2 according to EDS analysis is 89.8 wt.% Cu and 10.2 wt.% Sn (three measurements of ground areas).

ICP-MS analysis of coin no. 5.1 revealed that it was made of a ternary Cu-Sn-Pb alloy, with composition of 69.9 wt.% Cu, 4.3 wt.% Sn, and 24.8 wt.% Pb as the main alloying elements. Minor amounts of the Sb and As, and traces of the elements Ag, Te, Hg, Bi, Au, P, S, Co, Zn, and Se, were also spotted (Table 4). In contrast, ICP-MS examination of Coin no. 5.2 revealed that it was made of a binary Cu-Sn. Pb, Ni

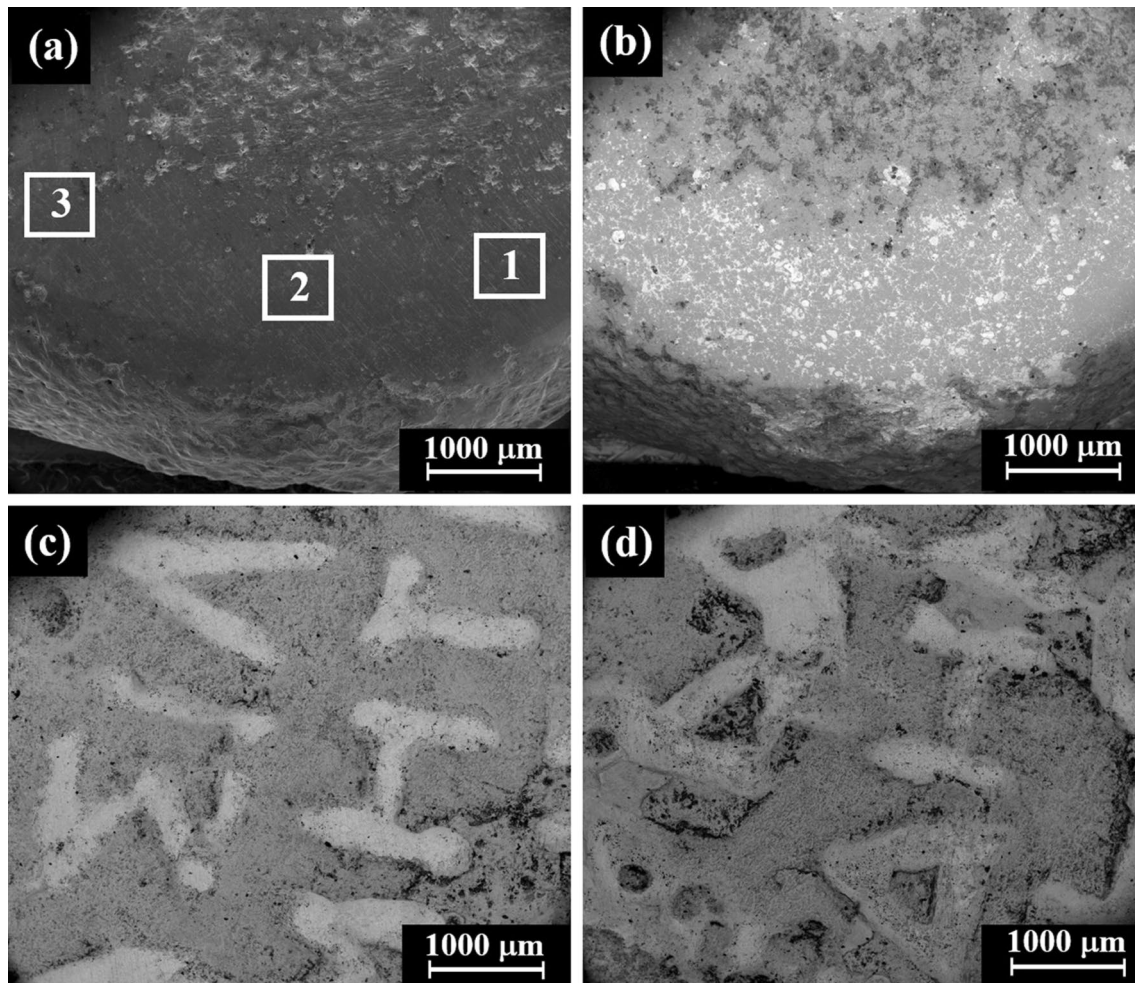


Fig. 11 SEM images of Group V's samples, reverse side after grinding the surface: (a, b) general view of the lower part of coin no. 5.1 (SE and BSE mode, respectively), where areas 1–3 were examined by

EDS; and (c, d) coin no. 5.2, struck letters, parts from the words *ΒΑΣΙΛΕ[ΩΣ]* and *ΔΗΜΗΤΡΙΟΥ* (BSE mode)

and As, and traces of the elements Ag, Sb, Te, Hg, Bi, Au, P, S, Co, Zn, and Se, were also detected (Table 4).

Test Case Coin Nos. T1 = SC 2119.2 (mint of Aké-Ptolemaïs), T2, T3 = SC 1672.1–2 (mint of Tyre)

SEM observation of coin no. T1 (Fig. 1) shows a rough surface of the patina and mud cover (Fig. 13a and b, SE and BSE, respectively, right and central part of the image). The ground surface near the circumference of coin T1 revealed dark areas rich in copper and brightislands rich in lead according to BSE mode (Fig. 13c). SEM observation of coin no. T3's patina and mud cover (Fig. 1) also revealed rough surface (Fig. 13d, BSE mode, right side of image).

The test case coin nos. T1, T2, T3 from Tell Izṭabba were all made of ternary copper-tin-lead alloy (Table 3). EDS analysis results of these coins alloy composition revealed an

average alloy composition of 71.3 ± 12.9 wt.% Cu, 8.3 ± 3.2 wt.% Sn, and 20.4 ± 14.8 wt.% Pb (nine measurements of ground areas).

XRD analysis results of the green patina and hard mud powder of coin T1 (Fig. 14a) revealed the presence of the minerals: cerussite, PbCO_3 ; diopside, $\text{CaMgSi}_2\text{O}_6$; clinocllore1MIlb, $(\text{Mg,Al,Fe})_6(\text{Si,Al})_4\text{O}_{10}(\text{OH})_8$; and clinoenstatite, MgSiO_3 . XRD analysis results of the green patina and hard mud cover of coins T2 (Fig. 14b) and T3 (Fig. 14c) revealed the presence of the minerals: calcite, $\text{Ca}(\text{CO}_3)$; quartz, SiO_2 ; plumboferrite, PbFe_4O_7 . In addition, coin T2 revealed the presence of montmorillonite, $(\text{Na})_{0.33}(\text{Al,Mg})_2(\text{Si}_4\text{O}_{10})(\text{OH})_2 \cdot n\text{H}_2\text{O}$; and coin T3 revealed the presence of phlogopite, $\text{KMg}_3\text{AlSi}_3\text{O}_{10}(\text{F,OH})_2$; and zincochromite, ZnCr_2O_4 . XRD analysis results of the thin brown layer of coins T1, T2, T3 (Fig. 15) after the removal of the green patina and hard mud cover revealed the presence of: copper; lead; iron-nickel-tin intermetallic compound,

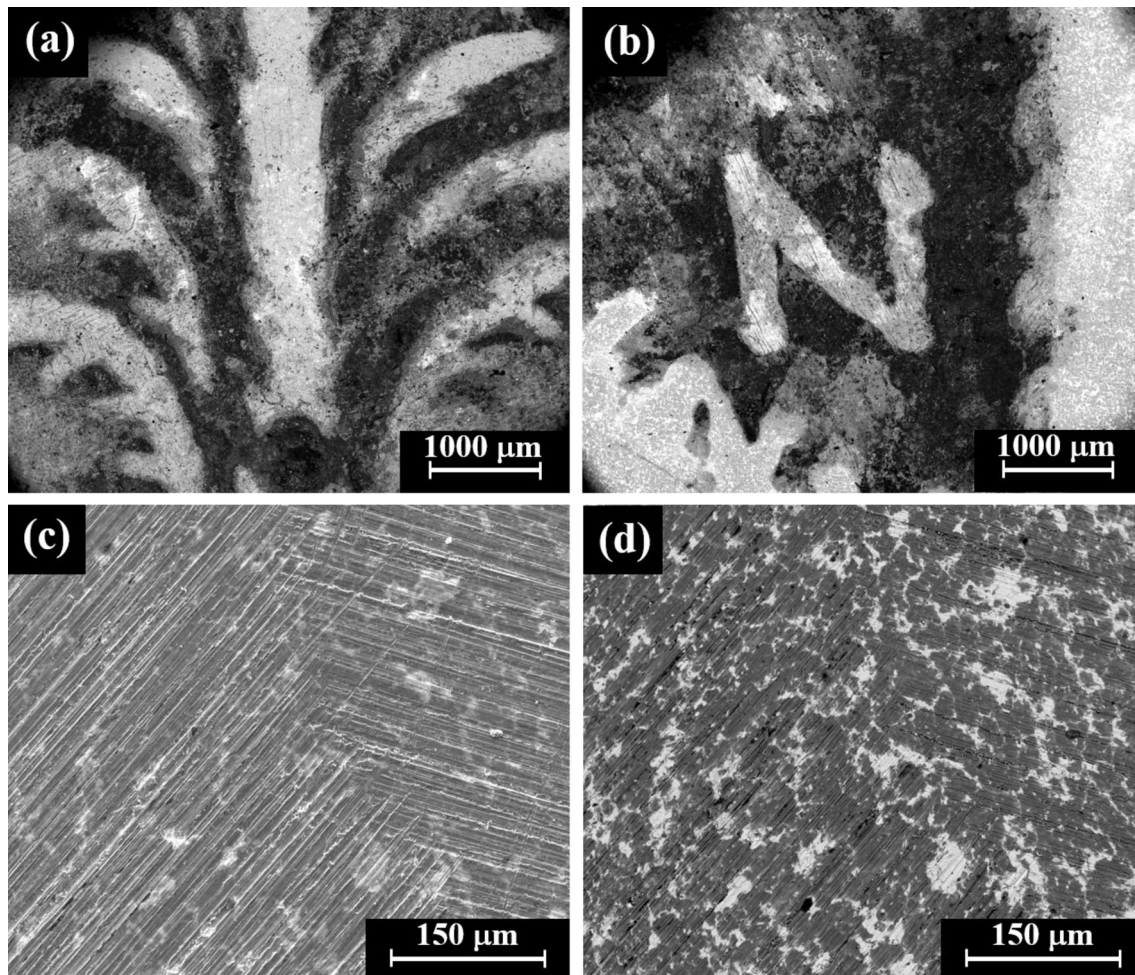


Fig. 12 SEM images of coin no. 5.3, Group V, reverse side after grinding the surface: (a) the struck palm tree branches (BSE mode); (b) the struck letter “N” at the lower part of the coin, where

NP represents the date 154 BSE; (c, d) the examined area 1 near the circumference of the coin

(Fe, Ni)Sn₂; cotunnite, PbCl₂; and zircon, Zr(SiO₄), in the case of coin T1; copper and copper-lead chloride hydroxide, CuPb₂Cl₃(OH)₂, in the case of coin T2; and copper-cobalt-iron-nickel-titanium intermetallic compound, Cu_{0.238}Ti_{0.048}Fe_{0.238}Co_{0.238}Ni_{0.238}; nickel-zinc intermetallic compound, NiZn; copper-zinc intermetallic compound, Cu_{0.7025}Zn_{0.2975}; and cuprite, Cu₂O, in the case of coin T3. It should be noted that XRD detection of phases containing trace elements such as tin may be limited when these elements occur at very low concentrations; therefore, such phase identifications are based on the closest matches to the diffraction patterns and should be interpreted with caution.

Lead Isotope Analysis

The LIA results of coins 1.2, 2.6, 3.2, 4.1, 5.1, and 5.2 are plotted for both ²⁰⁶Pb/²⁰⁴Pb vs. ²⁰⁷Pb/²⁰⁴Pb (Fig. 16a) and ²⁰⁶Pb/²⁰⁴Pb vs. ²⁰⁸Pb/²⁰⁴Pb (Fig. 16b) isotope ratios [57]. An

overlap was observed between Cyprus and Anatolia as possible ore sources of the lead raw metal observed, as shown in the ²⁰⁸Pb/²⁰⁴Pb vs. ²⁰⁶Pb/²⁰⁴Pb diagram (Fig. 16b).

Discussion

In the current study, five groups of Seleucid bronze coins recovered from the Tell Izṭabba, dated to the second century BCE, were studied by metallurgical analysis in order to shed light on their composition, microstructure, manufacturing process, and if possible, their metal origin. Such metallurgical analysis can provide important information on minting and technological skills, economic trends and constraints, political considerations, numismatic and cultural interactions, and transformations between ancient groups and societies [9, 21, 48, 58].

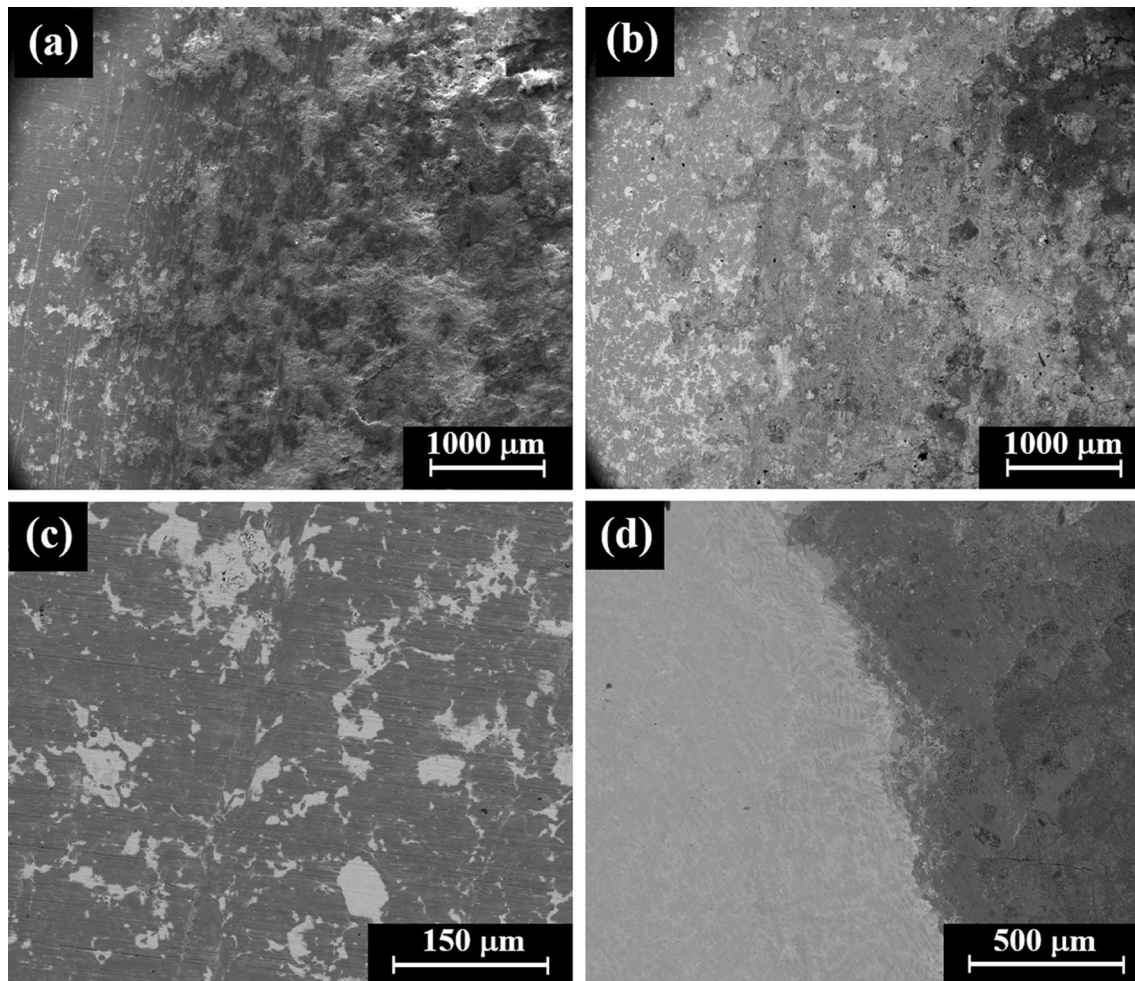


Fig. 13 SEM images of the test coins: **(a, b)** coin no. T1, where the area at the left side of the image was ground and the right area of the image remained covered with green patina and hard mud coating (SE and BSE mode, respectively); **(c)** higher magnification of the upper

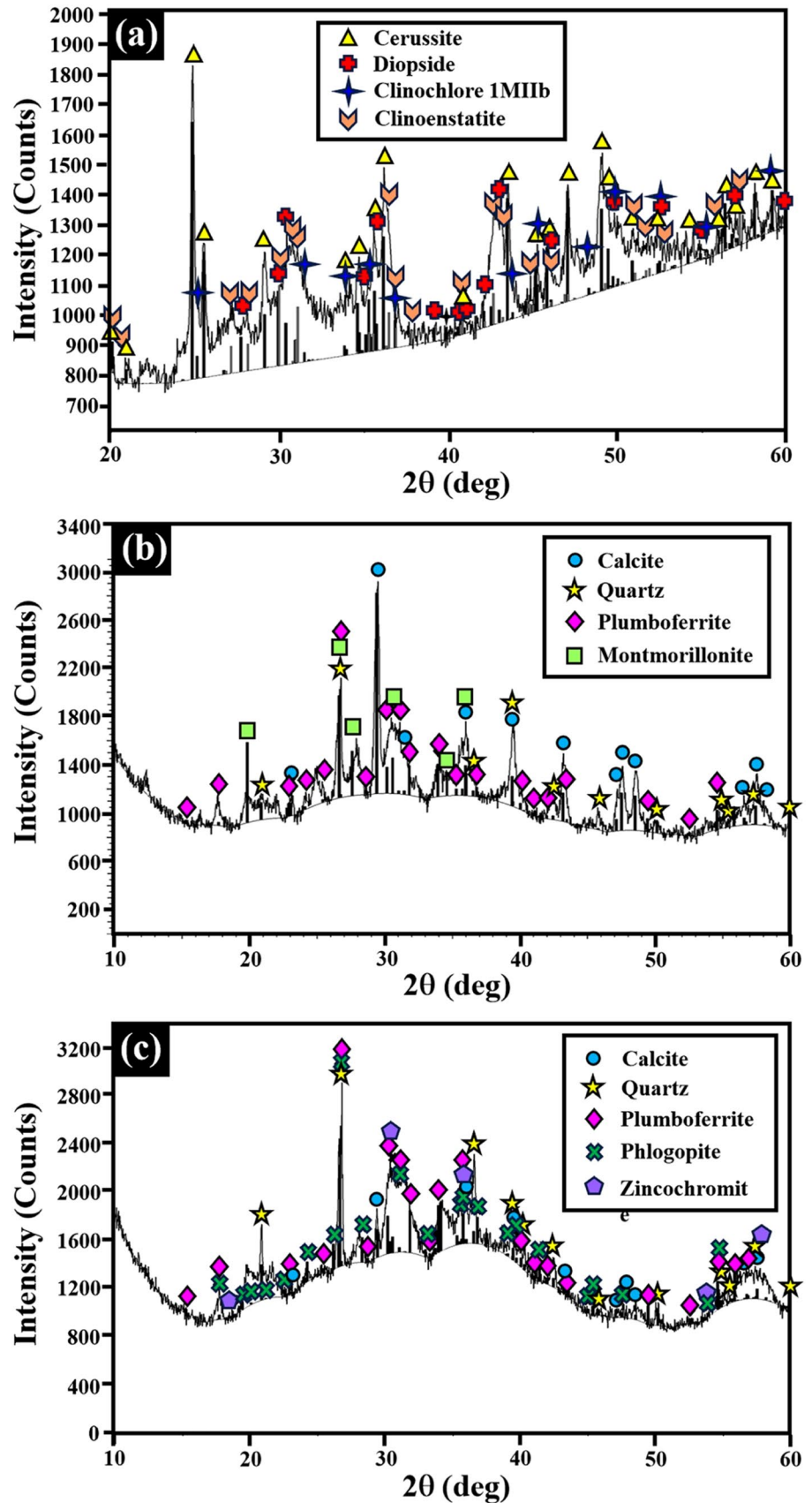
left side of the coin after polishing, examined area 1; and **(d)** coin no. T3, where the area at the left side of the image was ground and the right side remained covered with patina and hard mud coating

Determining the alloy composition of ancient bronze coins is quite challenging due to a rather thick patina layer with broadly varying concentrations of the main components [31]. In the present study, conventional destructive metallography was not feasible due to the fair state of preservation of the coins, which retain sharp strike details, and the limited number of specimens representing each compositional group. To overcome this limitation, we combined multiple analytical approaches to obtain a comprehensive understanding of the alloy composition and microstructural features, where the analytical strategy was aimed to minimize damage while obtaining the maximum amount of information. To reduce surface effects, the coins were cleaned by sandblasting before performing the XRF analysis, and then several areas on both sides of each coin were carefully ground prior to SEM-EDS analysis to reveal their metal core. In addition, one representative coin from each group was sampled

by drilling and removing a small amount of material from the coins' core for ICP-MS and lead isotope analysis. By integrating these complementary methods, we were able to account for surface corrosion and compositional heterogeneity, providing insight into internal lead distribution and production-related features without the need for destructive sectioning. The metallurgical analysis results of the five groups of Hellenistic coins (19 coins), as well as the three test coins, retrieved from Tell Iztabba, revealed that they are all made of ternary copper-tin-lead alloy, with the exception of coin no. 5.2 (Group V) that is made of a binary Cu-Sn alloy.

A heterogeneous distribution of the coins' components of the alloy (Cu, Sn, Pb) was observed by both XRF (Tables 2 and 5) and SEM-EDS analyses (Table 3 and 6). Ternary composition diagrams provide an effective way to visualize compositional relationships within Cu-Sn-Pb

Fig. 14 XRD analysis results of the green patina and hard mud cover of: **(a)** coin no. T1's patina powder, detecting the presence of the minerals cerussite (PbCO_3), diopside ($\text{CaMgSi}_2\text{O}_6$), Clinocllore 1MIb, $[(\text{Mg}, \text{Al}, \text{Fe})_6(\text{Si}, \text{Al})_4\text{O}_{10}(\text{OH})_8]$, Clinoenstatite (MgSiO_3); **(b)** coin no. T2, detecting the presence of the minerals calcite [$\text{Ca}(\text{CO}_3)$], quartz (SiO_2), plumbosferrite (PbFe_4O_7), and montmorillonite $[(\text{Na})_{0.33}(\text{Al}, \text{Mg})_2(\text{Si}_4\text{O}_{10})(\text{OH})_2 \cdot n\text{H}_2\text{O}]$; and **(c)** coin no. T3, detecting the presence of the minerals calcite, quartz, plumbosferrite, phlogopite [$\text{KMg}_3\text{AlSi}_3\text{O}_{10}(\text{F}, \text{OH})_2$], and zincochromite (ZnCr_2O_4)



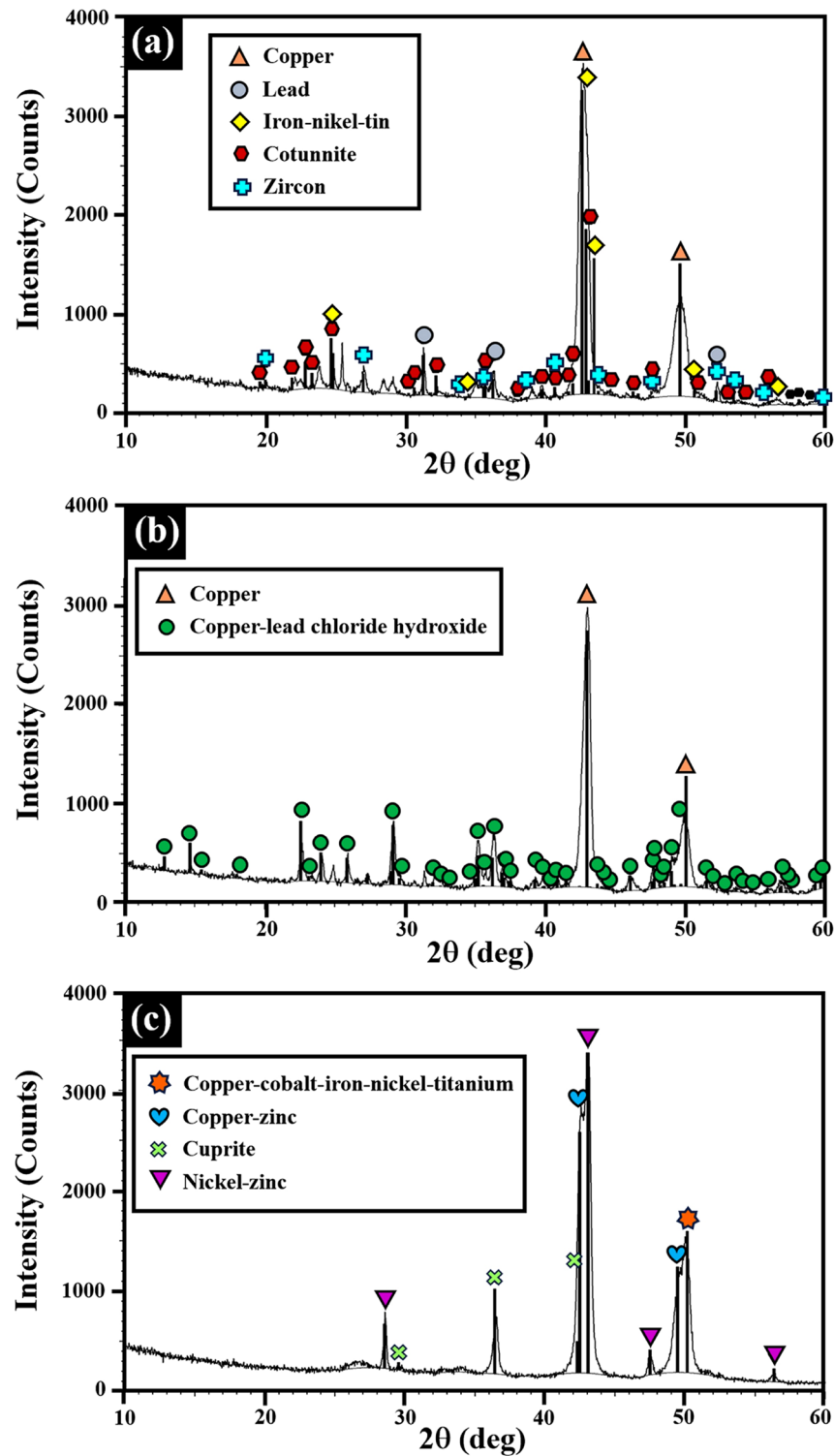
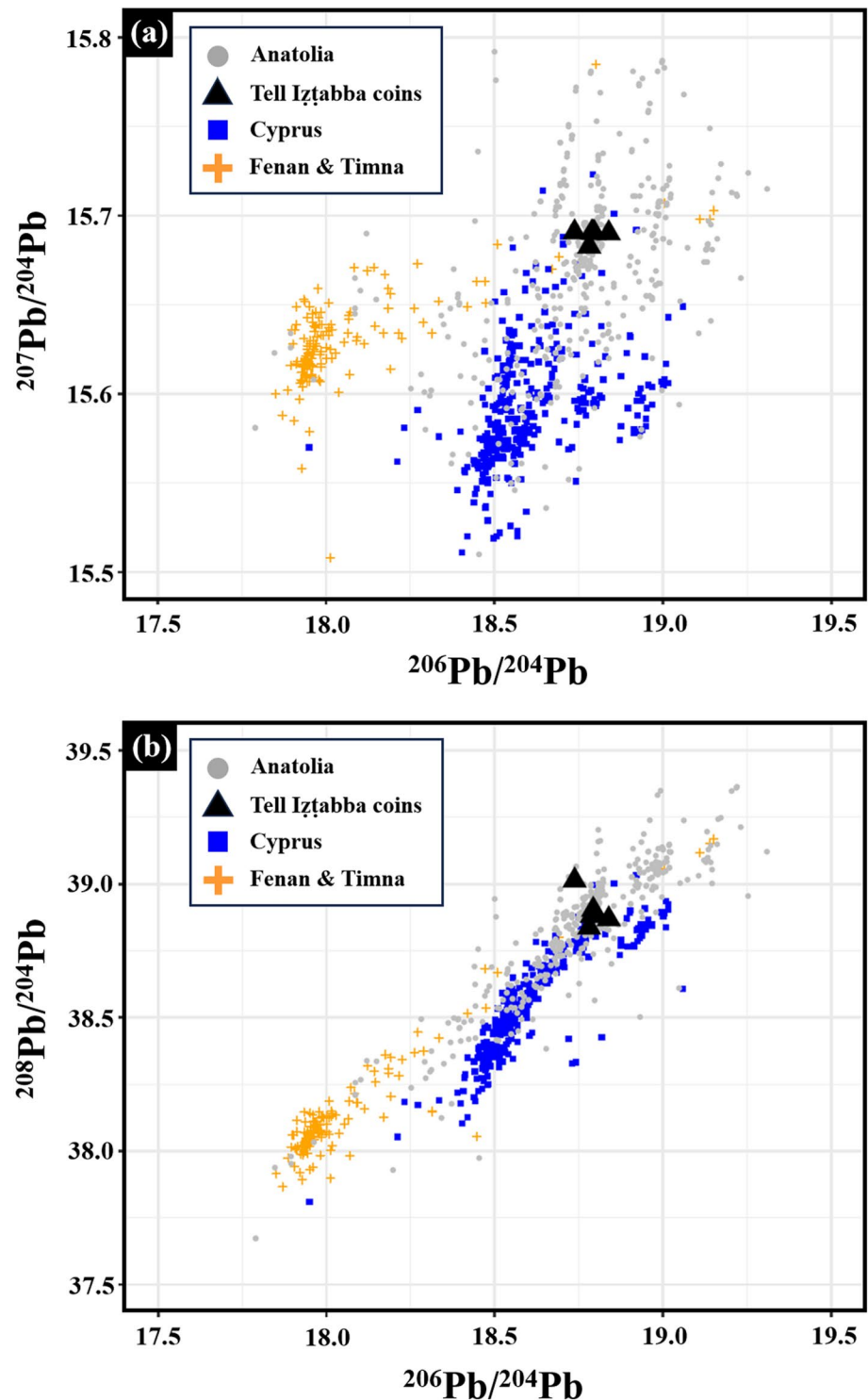


Fig. 15 XRD analysis results of the thin brown layer after the removal of the green patina and hard mud cover: **(a)** coin no. T1's surface, revealing presence of copper, lead, iron-nikel-tin [(Fe, Ni)Sn₂] intermetallic compound, cotunnite (PbCl₂), and zircon [Zr(SiO₄)] **(b)** coin no. T2's surface, revealing presence of copper and

copper-lead chloride hydroxide [CuPb₂Cl₃(OH)₂]; and **(c)** coin no. T3's surface, revealing presence of copper-cobalt-iron-nickel-titanium (Cu_{0.238}Ti_{0.048}Fe_{0.238}Co_{0.238}Ni_{0.238}) intermetallic compound, copper-zinc (Cu_{0.7025}Zn_{0.2975}) intermetallic compound, cuprite (Cu₂O), and nickel-zinc (NiZn) intermetallic compound

Fig. 16 Lead isotope ratio diagrams: (a) $^{206}\text{Pb}/^{204}\text{Pb}$ vs. $^{207}\text{Pb}/^{204}\text{Pb}$; and (b) $^{206}\text{Pb}/^{204}\text{Pb}$ vs. $^{208}\text{Pb}/^{204}\text{Pb}$. Results of coin nos. 1.2, 2.6, 3.2, 4.1, 5.1, and 5.2 are compared with published lead isotope reference data of ores from Fenan and Timna, Anatolia and Cyprus



alloys. By plotting the relative proportions of the three components simultaneously, these diagrams facilitate the identification of compositional clusters, technological trends, and potential differences between alloy groups [31]. The ternary diagrams of the bronze coins (Groups I–V) reveal significant compositional heterogeneity and a broad range of lead concentrations (Fig. 2, based on XRF

measurements), whereas the tin content varies within a comparatively narrow range. From a metallurgical standpoint, this pattern reflects the limited solubility of lead in the copper matrix and its tendency to segregate during solidification [23, 35, 36]. Among the groups, coin Group V displays the lowest lead content and the smallest compositional dispersion, while the greatest variability in

lead is observed in coin Group III. Despite this variability, the overall compositional ranges partially overlap between the groups (Fig. 17). Although the compositions display considerable heterogeneity, particularly in lead content,

the coins consistently fall within the Cu-Sn-Pb bronze system, and partial overlap between the compositional ranges of the different groups is observed. This pattern suggests a degree of alloying standardization and broadly similar

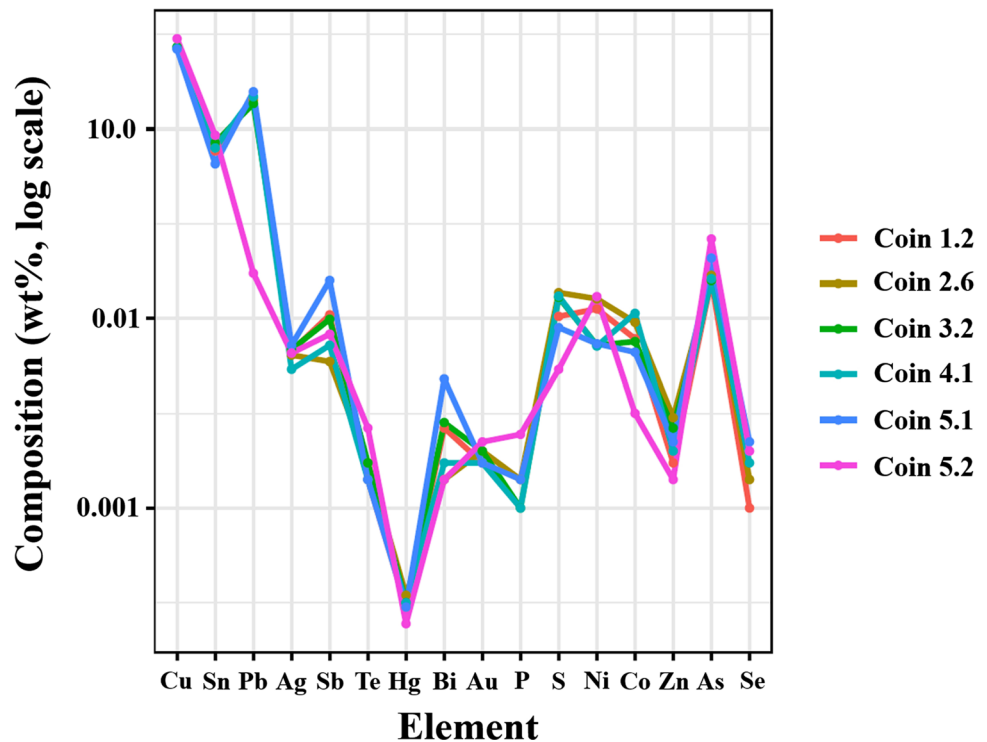
Table 5 The average alloy composition (wt.% content), according to XRF analysis, of the different groups of coins (Groups I–V)

Group of coins	No. of examined coins	No. of XRF measurements	Average Cu content (wt.%)	Average Sn content (wt.%)	Average Pb content (wt.%)
I (Nysa)	4	12	71.9±9.1	9.5±1.9	18.6±7.5
II (Antioch on the Orontes)	6	18	69.9±7.1	9.7±3.3	20.4±8.1
III (Antioch on the Orontes)	4	15	65.8±9.8	9.9±1.6	24.3±9.8
IV (Antioch on the Orontes)	2	6	72.1±9.0	8.7±1.1	15.4±8.6
V (Aké-Ptolemaïs)	2	6	76.4±4.1	7.6±0.7	16.1±4.5
V, exceptional coin 5.2	1	3	88.9	10.4	0.7

Table 6 The average alloy composition (wt.% content), according to SEM-EDS analysis, of the different groups of coins (Groups I–V)

Group of coins	No. of examined coins	No. of EDS measurements	Average copper content (wt.%)	Average tin content (wt.%)	Average lead content (wt.%)
I (Nysa)	4	12	64.0±8.4	10.5±3.5	25.5±10.3
II (Antioch on the Orontes)	6	19	66.7±15.9	12.2±6.5	21.1±14.3
III (Antioch on the Orontes)	4	13	66.9±10.1	7.4±1.9	25.7±10.0
IV (Antioch on the Orontes)	2	6	72.2±7.6	8.3±1.0	19.5±6.9
V (Aké-Ptolemaïs)	2	6	71.8±6.9	6.7±1.1	21.5±7.5
V, exceptional coin 5.2	1	3	89.8	10.2	–
T (unknown mint)	3	9	71.3±12.9	8.3±3.2	20.4±14.8

Fig. 17 Ternary composition (wt.%) diagram of Cu-Sn-Pb of the coins (Groups I–V) from Tell Izṭabba based on XRF analysis: (a) full compositional range; and (b) enlarged view of the data cluster



metallurgical practices across the groups and possibly a shared or closely related source of raw materials.

In BSE-SEM imaging, areas that appear brighter correspond to regions containing elements with higher atomic numbers, while darker areas indicate regions richer in lighter elements. Therefore, the contrast observed on the coins' ground surfaces reflects the relative distribution of lead and copper. Although the bulk composition was heterogeneous, consisting of bright islands rich in lead (Pb globules) and a dark copper–tin matrix, as observed in SEM BSE mode (Figs. 3, 4a, 5, 7a–b, 7e–f, 8b–c, 8f, 9b–d, 10, 11b, 12d, 13b–c), all the coins (except coin no. 5.2) were nevertheless produced from copper–tin–lead alloys with similar average compositions. This consistency, despite originating from different mints (Antioch on the Orontes, Aké-Ptolemaïs and Nysa-Scythopolis), suggests a certain degree of standardization. SEM-EDS elemental mapping (coin 1.2, Fig. 4a) revealed that areas rich in copper (Fig. 4b) also contain a higher amount of tin (Fig. 4d), whereas areas rich in lead (Fig. 4c) contain a higher amount of chlorine (Fig. 4f), according to the formation of PbCl_2 and $\text{CuPb}_2\text{Cl}_3(\text{OH})_2$. SEM-EDS spectrum of coin nos. 2.4 and 3.2 also supports this correlation between Pb and Cl (Fig. 6). The corrosion products of the bronze coins (Figs. 14 and 15) reveal different traces of chemical interactions between the metallic substrate and its environment through time and therefore can be considered as pertinent historical evidence (corrosion biography of the item). Such information can be useful for the conservation of such coins [35, 36].

The high concentration of lead in the coins was probably related to technological considerations of increasing the fluidity and castability of the metal [10, 23, 30]. Yet, for optimal fluidity through casting, only 2 wt.% Pb was needed [59]. Since lead was considerably cheaper than copper and tin in antiquity, it was frequently added to copper alloy coins for economic reasons [23, 32, 36]. Therefore, the use of high lead concentrations in the current Hellenistic bronze coins probably also links to economic constraints. Yet, a high lead content was probably also introduced to enhance the metal's malleability and improve the striking quality during the coin minting process [31].

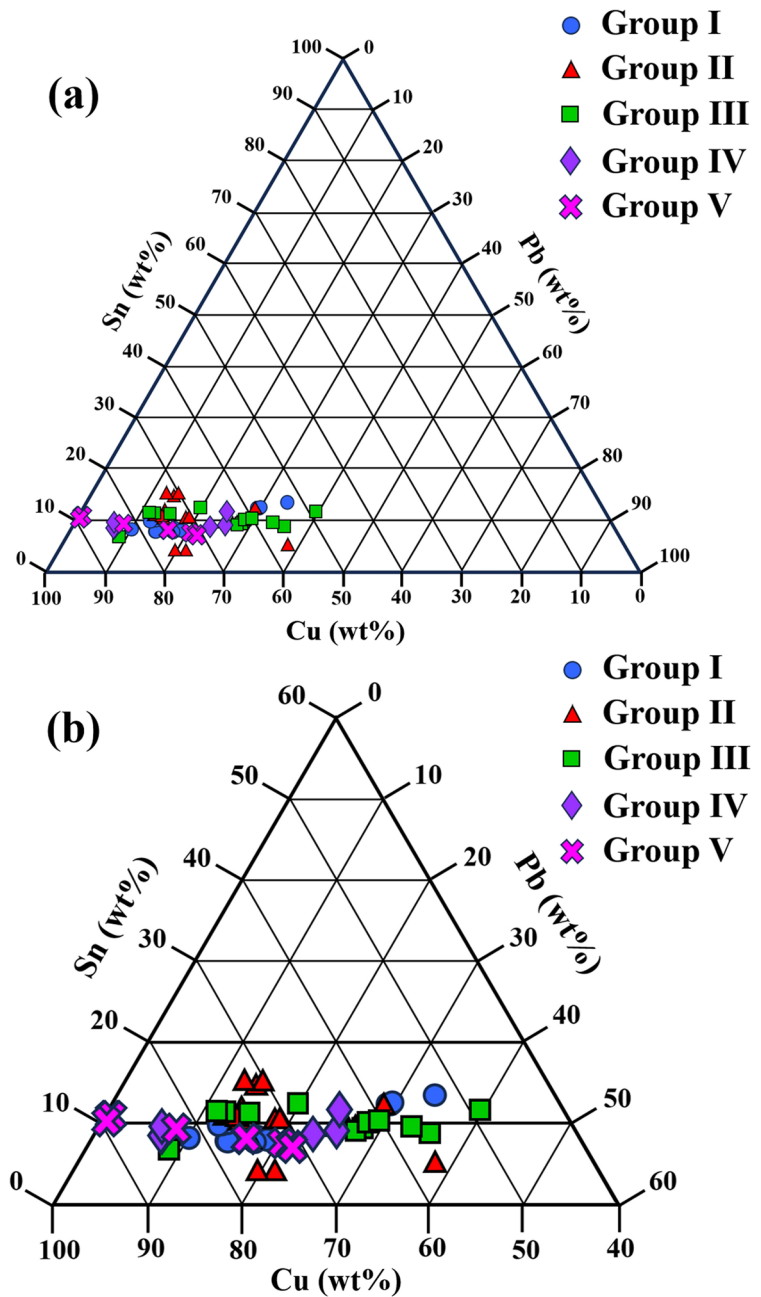
While bronzes have good corrosion resistance, ancient bronze items tend to interact with their surroundings through time, thus suffering from corrosion damage [35]. The presence of plumboferrite mineral on the external green patina and hard mud cover of bronze coins (Fig. 14, coin nos. T2 and T3) and of lead and lead compounds, such as cotunnite (Fig. 15a, coin T1), and copper–lead chloride hydroxide (Fig. 15b, coin T2) on the thin brown layer after the removal of the green patina and hard mud cover is probably associated with segregation or corrosion processes [36, 60]. XRD analysis results of the green patina and hard mud cover revealed lead compounds but not copper compounds

(Fig. 14), yet copper and copper compounds were detected in the thin brown layer after the removal of the green patina (Fig. 15). Throughout the patination process, copper is released from the object's surface into the surroundings, and lead and tin tend to accumulate there instead. As a result, the copper content in the external surface decreases between 20 and 60 wt.% Cu, whereas the lead and tin concentrations are up to four times higher than in the bulk alloy [31, 61].

The Cu–Sn–Pb coins recovered from Tell Iztabba were produced in diverse mints based on their reading (Groups I–V). However, the ICP-MS analysis results reveal a highly consistent of minor and trace element pattern across all examined coins' core, including traces of the elements Ag, Sb, Te, Hg, Bi, Au, P, S, Ni, Co, Zn, As, and Se (coin nos. 1.2, 2.6, 3.2, 4.1, 5.1, and 5.2, Table 4 and Fig. 18), suggesting that the raw material used for the production of the coins from Groups I–V originated from the same or a closely related source. Elements such as As, Sb, Ni, Co, Ag, Se, S, and Te are often associated with the origin of the copper ore [48, 62–64], whereas copper derived from oxidized ores exhibits enrichment in Ni and Ag, accompanied by the presence of As and Sb [65]. Elements such as Ag, Bi, Sb, As, and occasionally Au, Hg, Cu, Zn, and Se, point to the lead ore [66–69]. For a comparison, the minor and trace element composition of copper oxide ore from Timna revealed the presence of Fe, Co, Ni, Mn, Ag, Zn, As, and S. Smelting experiments conducted by Tylecote et al. (1977) [65] on Timna ore further showed the presence of Fe, Co, Ni, Mn, Ag, Zn, As, S, Sn, Pb, Sb, and Bi as minor and trace elements. Similarly, Roman copper alloy objects from Timna and neighbouring sites contained Fe, Co, Ni, Mn, Ag, Zn, As, Sn, Pb, Sb, and Bi as minor and trace elements [65]. Although the microstructure and composition of each of the coins are heterogeneous, the overall chemical similarity between the coins observed in the ICP-MS results (Fig. 18) suggests that the coins were likely produced within a unified metallurgical tradition or under centralized control of metal supply, reflecting both technological uniformity and economic or administrative hegemony in the region.

Based on VT inspection and SEM observation, it is suggested that manufacturing process of the coins included three steps: (1) flans were cast, (2) the flans were then probably heated to relieve residual stresses and restore the ductility of the alloy (this interpretation is supported by the good preservation of the examined coins, which display sharply defined motifs without visible cracking or tearing of the metal), where in some cases, the flans were also ground prior to striking, (3) obverse and reverse dies were used to strike the motifs [12, 36, 58, 70]. Some of the coins have angular cut marks (Fig. 1) that indicate that they were cast into a mould with concavities, either with a strip shape or with a tree shape, which would then detached/cut into isolated flans [35, 58, 71–73]. Worn or damaged dies were used at times

Fig. 18 ICP-MS analysis results of the dissolved powder drilled from the coins' bulk: The elemental composition of coin nos. 1.2, 2.6, 3.2, 4.1, 5.1, and 5.2 [discontinued lines = values below detection limit]



when there was a shortage of skilled engravers [58, 74], as the reverse die (also named the punch die), located on the hammer side, tends to deteriorate before the lower obverse die (also named the anvil die) [58, 75, 76].

In the present study, XRF, SEM-EDS, and ICP-MS analyses were employed to characterize the composition and heterogeneity of the coins, to gain insights into their manufacturing technology, and to explore information concerning the provenance of the coins' raw material. All three methods revealed that the Tell Iztabba coins were made of Cu-Sn-Pb alloy, with quite similar results for major elements in all three methods. Yet, each technique

offers distinct advantages and limitations, and the selection of these methods was related to the current study research objectives as well as the intention to preserve the coins as much as possible. XRF chemical analysis was performed since it is a rapid, non-destructive testing (NDT) method that enables determination of major and minor alloying elements [77, 78]. Its portability and minimal sample preparation made it particularly suitable for initial analysis. However, XRF is limited by its sensitivity to surface conditions, where corrosion layers, patinas, and heterogeneous oxidation can lead to elemental enrichment or depletion at the surface, thereby influencing the analytical

results. Consequently, XRF data are often considered representative of surface composition rather than the bulk alloy [77, 78]. SEM-EDS NDT analysis was applied for the examination of the coins since it offers high resolution as well as establishing a correlation between the microstructure and the elemental composition of the coins [62, 74]. This technique allows visualization of metallurgical features such as corrosion products, defects, and phase segregation, which are critical for understanding of the manufacturing process [36, 79]. Nevertheless, SEM-EDS is semi-quantitative and less accurate for elements with low atomic numbers or those present in trace amounts (below ~0.1 wt.%). Moreover, similar to XRF results, SEM-EDS results can be influenced by surface corrosion and require sample preparation, often involving polishing or preparation of metallographic samples [80]. ICP-MS analysis, by contrast, provides highly precise and quantitative elemental analysis across a broad range of elements, including major elements, minor elements, and trace elements [63, 81]. This technique requires complete dissolution of the sample, making it destructive [81], but it yields detailed chemical fingerprints that can be used to assume the provenance of raw materials and identify recycling or mixing processes [62]. In the case of examined coins, the ICP-MS enabled reliable quantification of trace elements such as Ag, As, Sb, Bi, and Ni, which serve as geochemical indicators [48, 62–64, 66–69].

The analysed Seleucid coins from Tell Izṭabba were made of a ternary copper-tin-lead alloy and exhibited a heterogeneous microstructure according to SEM observation (Figs. 3–13). Only one coin (no. 5.2, Fig. 11c–d) was composed of a binary copper-tin alloy. Based on these findings, one of our main research questions is whether the flans used for minting were produced in the same mint that struck the coins, or in a regional workshop supplying flans to several mints. Based on the current XRF, EDS, and ICP results (Tables 2, 3, 4, 5 and 6) combined with the coins' typology, it seems that flans were produced in the mint that struck them, despite the (common) high concentrations of lead that were added to the coins' alloy of all groups. Moreover, given the LIA plots (Fig. 16), the analysed coins exhibit a homogeneous lead isotope signature (Fig. 16), indicating that the Levantine mints (Antioch, Aké-Ptolemaïs, and Nysa-Scythopolis) probably used metal derived from the same ores, that may have been because of their Seleucid imperial control at the time. As for the mineral sources, Fenan and Timna ores can be excluded due to differing lead isotope signature. Lead isotope overlap exists for Cyprus and Anatolia ores hindering a definitive decision between these two mineral source regions. The addition of lead to the coins also complicates the application of the lead isotope method, as copper and lead may have different origins, and lead isotope mixing could result from alloying.

Conclusions

The analysed Hellenistic bronze coins (Groups I–V) from Tell Izṭabba are made of ternary copper-tin-lead alloy with one exception (no. 5.2) that is made of binary copper-tin alloy (with only 0.7 wt.% Pb and 0.3 wt.% Pb as impurity element according to XRF analysis and ICP-MS analysis, respectively, and no presence of Pb according to EDS analysis). The ternary diagrams of the bronze coins (Groups I–V, XRF measurements) reveal considerable compositional heterogeneity, particularly in lead content, whereas tin varies within a comparatively narrow range. Despite this variability, the compositional fields of the groups partially overlap, and most samples cluster within a relatively restricted region of the diagram. This pattern suggests a degree of alloying standardization and broadly similar metallurgical practices, possibly indicating a shared or closely related source of raw materials.

The different metallurgical analyses results demonstrate some level of standardization in the production of the coins, even though the coins were produced in different mints. ICP-MS results reveal close similarity in the minor and trace element composition of all examined coins, suggesting that the raw metal originated from a single or closely related source. Although the flans were almost certainly produced in the same mints that struck the coins, the metallurgical data do not support the existence of mint-specific alloy “recipes”. On the contrary, the XRF, SEM-EDS, ICP-MS, and LIA results show that coins from different mints share broadly similar Cu-Sn-Pb compositions, comparable distributions of minor and trace elements, and overlapping lead isotope signatures. This consistency indicates that the south Levantine mints operated within a shared metallurgical tradition and drew upon common or closely related metal sources, rather than employing individualized or locally distinctive flan-preparation formulas.

The consistently high lead content in the alloys was likely the result of technological and economic considerations. LIA data, showing isotopic overlap between Cyprus and Anatolia, further suggest that Levantine mints utilized similar ore sources even under Seleucid rule. This pattern reflects a well-organized network of metal procurement and distribution across the eastern Mediterranean, possibly coordinated or influenced by the centralized Seleucid administration.

The technological uniformity and shared metallurgical practices evident in the coins point to both economic integration and the transmission of metallurgical knowledge within the region.

Overall, this research offers new insights into the technological minting skills and economic aspects of bronze coins production in the southern Levant during the

Hellenistic period, thereby enhancing our understanding of the material culture during the period.

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